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Hybrid deep learning framework for cardiovascular risk prediction in post-COVID-19 patients

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Cardiovascular complications associated with Post-COVID-19 Patients remain difficult to identify at early stages due to heterogeneous physiological manifestations and the limited integration of imaging and clinical indicators in conventional diagnostic frameworks. To address this challenge, this work proposes a multimodal deep learning framework for cardiovascular disease (CVD) risk prediction by integrating cardiac computed tomography (CT) imaging with structured clinical data. The proposed approach applies Adaptive Bilateral Filtering for image enhancement, Kernel Density Fuzzy C-Means (KDFCM) for myocardial segmentation, and Squeezing Extract Chirplet Transform (SSECT) for extracting multi-scale texture and frequency-based imaging features. Clinical variables are preprocessed through Z-score normalization, interquartile range (IQR)-based outlier removal, and Isolation Forest anomaly detection, followed by Adaptive Starfish Optimization (ASO) for feature selection. Imaging and clinical representations are integrated through feature-level fusion and classified using a Deep Image Recognition–based Generative Adversarial Network (DIR-GAN), where adversarial learning enhances discriminative feature representation and model robustness. Experimental evaluation demonstrates strong predictive performance, achieving 94.98% accuracy, 94.98% sensitivity, 93.38% specificity, 95.43% precision, and 94.74% F1-score. The findings indicate that the proposed framework provides reliable and clinically relevant cardiovascular risk stratification for patients recovering from COVID-19, while demonstrating robustness against heterogeneous multimodal data.

Keywords Cardiovascular, COVID, Deep learning, Preprocessing, Bilateral Filtering, Fuzzy C-means clustering, Starfish Optimization

Cardiovascular disease is the primary cause of death worldwide and encompasses a variety of diseases involving the heart and blood vessels, such as coronary artery disease, congestive heart failure, cardiac arrhythmia, and cerebrovascular stroke. It has a worldwide impact on both developed and underdeveloped nations, so it seems to affect everyone equally. In the United States, cardiovascular disease affects over 655,000 people each year. These diseases are also responsible for almost 28.1% of all deaths in countries, such as India. The data reported by the World Health Organization in 2019 showed that the majority of deaths worldwide were linked to cardiovascular diseases and that an effective diagnostic system and timely intervention are of great importance[1,2].

Although considerable breakthroughs have been achieved in medical diagnosis and therapeutic interventions, the problem of detecting cardiovascular disease in its early and asymptomatic phases continues to be a challenge, especially in areas with poor healthcare facilities. Standard clinical equipment and trained medical personnel are typically required for the use of standard clinical tools, including electrocardiography (electrography in which the heart's electrical pattern is recorded) and cardiac auscultation (auscultation in which the heart sounds are heard using a stethoscope). In isolated or marginalized locations, these requirements may restrict access. Cardiovascular assessment is more important now because of the outbreak of Coronavirus Disease 2019 (COVID-19) worldwide. Patients with pre-existing heart disease have a higher risk of complications (including ARDS, myocardial damage, and death). In patients with cardiovascular comorbidities, lactate dehydrogenase (LDH) has been demonstrated to be a predictive marker for poor outcomes in patients with COVID-19[3].

In this context, Artificial Intelligence (AI) and Machine Learning (ML) have emerged as promising approaches for addressing these diagnostic gaps. These types of calculations can be very useful in analyzing large structured

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medical datasets, such as patient age, cholesterol concentration, systolic and diastolic blood pressure, glucose concentration, smoking, and body mass index. These algorithms can be used to model complex or non-linear relationships that may not be modeled with traditional statistical techniques and can enhance the power of risk scores and streamline the clinical workflow. Various studies have discussed that the combination of ensemble learning systems and deep learning neural architectures is more effective than the use of any single classifier in the diagnosis and prediction of cardiovascular diseases[4].

Ensemble learning techniques capable of adaptation and good generalization have been the focus of many studies, particularly model stacking. Classifiers are grouped together into a base-level individual (Random Forest, Extreme Gradient Boosting, Extremely Randomized Trees, etc.) and then passed to a higher-level individual (meta-learner) to make more powerful predictions. Several institutions, including Cleveland and Hungarian, Statlog, Long Beach Veterans Affairs, and Swiss, have released data to demonstrate the superiority of these stacked models in several different ways to measure performance. These include classification accuracy, precision, recall, specificity, F1 score, and area under the receiver operating characteristic curve. Moreover, the Recursive Feature Elimination technique has been proven to be a data dimensionality reduction and redundant predictor removal technique that further optimizes the learning process and enhances the reliability of the model [5].

Simultaneously, improvements in non-invasive diagnostic methods have significantly contributed to the availability of early evaluation of cardiovascular disease. In clinical facilities where standards are below par, the phonocardiographic method of acoustic recording of heart murmurs is an inexpensive alternative to electrocardiography. Recently, a portable acquisition system for phonocardiograms (PCG) was implemented using classifiers built using artificial neural networks trained to recognize abnormal heart sounds. The system was able to use signal characteristics, such as skewness, kurtosis, spectral slope, roll-off, and spread, to distinguish between normal and pathological cases. The classification results of the 4600 recordings with labels were nearly perfect, illustrating the ability of the intelligent and affordable screening tools to identify and classify at the community level[6].

The use of several data modalities has also been studied to enhance the diagnostic accuracy. An interesting configuration was implemented in which two convolutional neural networks (CNN) processed the electrocardiographic and phonocardiographic data streams, which were named the Cardiac-Lead Electrocardiogram Network and Cardiac-Lead Phonocardiogram Network. These deep architectures were juxtaposed with an optimal feature selection through a genetic algorithm and finally a support vector machine (SVM) for the final classification. This multimodal approach can outperform single-signal systems, reflecting the crucial role of integrative learning pipelines in complex clinical contexts [7].

To overcome the shortcomings of feature redundancy, class imbalance, and convergence to local optima, researchers have added metaheuristic optimization methods to machine learning models. The first approach was a sample-based implementation of a neural classifier combined with an algorithm that mimics the predatory behavior of hawks, called the Improved Hawks Optimizer. The hybrid approach was implemented on a semi-structured dataset with 1,025 patient records and 14 clinical parameters, which yielded better classification accuracy with acceptable efficiency in computation [8].

In Africa and other remote areas where medical facilities are scarce, such as sub-Saharan Africa, portable medical diagnostics, intelligent, smart, and scalable diagnostics are needed. To facilitate continuous monitoring and immediate generation of alerts, a Random Forest (RF) classifier was proposed based on the real-time data received from an IoT device equipped with wearable devices. The system has been developed to be deployed in remote areas such as Rwanda, where cardiovascular conditions are responsible for almost 12% of premature deaths among people aged 30–70 years, and help clinicians obtain actionable insights without the need for centralized healthcare services [9].

More advances have been directed towards making predictive models more interpretable and faster to process. For instance, a gradient-boosted decision tree and binary spotted hyena optimizer metaheuristic-inspired algorithm that has been used to simplify the feature selection process through hyperbolic tangent transformation has been found to be beneficial. It has been successfully applied to Heart Disease datasets from the UCI repository, resulting in better diagnosis accuracy and reduced computation time on that dataset in real-world healthcare applications [10]. The COVID-19 pandemic has reshaped the healthcare environment worldwide and revealed certain gaps in the management of chronic diseases, such as cardiovascular disease. Heart issues that persist or emerge after COVID-19 infection that are not detected by standard screening tests. To make this transition, predictive instruments are needed to interpret subtle clinical changes linked to post-viral cardiovascular stress. The rationale for this research is the immediate need for measures to provide intelligent healthcare systems with tools that are capable of detecting latent cardiac risk and are adaptable to the changing profile of patients affected by the long-term consequences of SARS-CoV-2 infection.

Despite the advances in machine learning approaches for CVD prediction, only a few have been adapted to reflect the differences in physiological changes that occur after COVID-19. Existing models are mostly based on static clinical parameters and do not consider inflammatory markers, tissue stress biomarkers, or systemic responses induced by previous viral exposure. Furthermore, the majority of current models are not flexible enough to consider dynamic changes in patient baselines during the months after infection. The current field of hybrid diagnostic architectures is far from complete, with no significant example of a specific architectural approach that has been created specifically to connect traditional CVD measures with subtle changes in clinical status that are occurring in the post-COVID-19 population.

- Introduced a multi-modal prediction system combining cardiac imaging and structured clinical data for robust CVD risk prediction in Post-COVID-19 patients.

- Developed an advanced image enhancement pipeline using Adaptive Bilateral Filtering to retain myocardial edges while suppressing noise.
- Proposed the use of KDFCM segmentation to handle fuzzy boundaries in CT scans and enhance myocardial delineation.
- Deployed SSECT-based feature extraction, enabling detection of localized frequency and texture anomalies associated with cardiovascular pathology.
- Implemented clinical data normalization, outlier elimination via IQR, and anomaly detection using Isolation Forest to ensure clean and balanced data.
- Applied Adaptive Starfish Optimization (ASO) for selecting the most informative clinical biomarkers, optimizing dimensionality and prediction relevance.
- Designed a feature-level fusion strategy with weighted scaling, integrating imaging and clinical representations into a unified diagnostic vector.
- Employed a DIR-GAN architecture for classification, boosting resistance to imbalanced datasets and improving generalization.

To conduct an extensive assessment of the proposed framework, the current research poses the following research questions for analysis purposes:

- RQ1: Is the combination of cardiac CT imaging and structured clinical data effective in improving the prediction performance of cardiovascular risk stratification in Post-COVID-19 individuals in comparison to conventional methods?
- RQ2: Is the proposed DIR-GAN framework better in terms of prediction, generalization, and robustness compared to traditional approaches to cardiovascular risk stratification?
- RQ3: What is the role of preprocessing, feature extraction, feature optimization, multimodal feature integration, and adversarial learning in predicting cardiovascular risk stratification using the proposed framework?

The remainder of the manuscript structured as follows. The next section examines the existing literature and summarizes the latest developments and ongoing challenges in the use of AI technologies for cardiovascular diagnostics. The methodology section provides a description of the hybrid design of the suggested model, including the data collection methods, feature engineering techniques, model architecture, and optimization strategies. The experimental setup is described in the next section, followed by a detailed description of the evaluation protocols and a discussion of the prediction performance on different benchmark datasets. Finally, the last part summarizes the main findings and suggests future research avenues, particularly in the clinical context, interpretation, and integration with novel remote monitoring tools.

Related works

Gupta et al.[11] used a stacking ensemble architecture based on deep neural networks and a Logit Boost meta-learner to determine the heart disease risk of people recovering from COVID-19. From an initial sample of 180 patients, their approach bootstrapped the data to add approximately 4,700 to the dataset, then filled in missing values with linear regression, and normalized the data to the min–max scale. The model was trained using cross-validation with 10 folds and outperformed the classic machine learning models across all folds. This work demonstrates the effectiveness of ensemble stacking in complementing small clinical datasets. Kumar et al.[12] split the training data into biased and regular sub-datasets to address the issue of data bias in a CHD prediction model. They used the Binary Krill Herd Algorithm for feature selection, Capsule Network with an existing Visual Geometry Group-16 (VGG-16) model for classification, and a Gaussian optimizer to optimize the network. They also added a Variational Autoencoder layer to reduce the overfitting problem and a Stacked Genetic Algorithm. Better generalization and bias mitigation strategies were also shown in the Cleveland heart disease dataset.

Khan et al.[13] designed a two-level diagnostic system, employing Phonocardiogram (PCG) signals acquired from a low cost custom-made stethoscope. They used statistical, spectral, and Mel-Frequency Cepstral Coefficients after the EMD preprocessing for feature extraction. A weighted K-Nearest Neighbor (KNN) classifier was used to classify normal and abnormal signals, whereas a cubic kernel Support Vector Machine (SVM-C) classified the disease severity as single, double, or triple vessel. This is a noninvasive, low-computation method that emphasizes the practicability of deployment. Nelson et al.[14] provided a solution called ABFog which brings fog computing for processing cardiopulmonary sensor data (e.g., electrocardiography (ECG), oxygen saturation, and temperature) close to the data source using the FogBus framework. In their solution, they employed an ensemble of tuned AlexNet networks, activated by rectified linear Unit (ReLU), with softmax outputs, which were used to achieve high accuracy in real-time classification while reducing latency and setting quality-of-service parameters.

Hasan et al.[15] designed a multi-stage feature selection method that preserved only features that were selected by a number of different methods (Chi-Square, Pearson correlation, Recursive Feature Elimination (RFE), Least Absolute Shrinkage and Selection Operator (LASSO), Random forest (RF), and Light gradient boosting machine (LightGBM)). They were trained on a large Kaggle dataset of 70,000 records and tested using classifiers such as Extreme Gradient Boosting (XGBoost), Support Vector Classification (SVC), KNN, Random Forest, Naïve Bayes, and found that RFE with boosting algorithms provides better prediction accuracy; however, it requires more computational time. Baghdadi et al.[16] used CatBoost along with Shapley value feature selection and tuned the model through hyperparameter optimization using Optuna. Key predictors were identified to provide clinically interpretable models with fewer diagnostic overheads, such as cholesterol, ST segment slope, blood pressure, age, and pain type.

Higaki et al.[17] found that a high NLR is an independent risk factor for mortality in a work on the neutrophil-to-lymphocyte ratio (NLR) as a prognostic biomarker in COVID-19 patients with cardiovascular comorbidities. Follow-up measurements, not baseline values, had greater predictive value for in-hospital mortality in the analysis of the CLAVIS-COVID registry, with an area under the receiver operating characteristic curve (AUC) of 0.893. Sai Krishna Reddy et al.[18] evaluated and compared two classifiers, Decision Tree (DT) and Gaussian Naïve Bayes (GNB), using the Heart Failure dataset from Kaggle (299 samples, 13 attributes). After removing less informative features, Gaussian Naïve Bayes outperformed it, particularly for small, continuous datasets, which it is suitable for, and thus can be used in resource-limited medical settings.

Sinha et al.[19] introduced a smart healthcare framework, iCardo, which is based on the Internet of Medical Things (IoMT). They created 16 feature sets using methods such as LASSO, RFE, Chi-Square, and tree-based feature selection, and trained seven models: SVM, Logistic Regression (LR), XGBoost, K-nearest neighbors, and ANN. The best results were achieved with an SVM on the RFE and Chi-square selected features, which can be integrated with IoMT platforms for real-time monitoring. Rahim et al.[20] presented a solution that uses a combination of preprocessing methods, such as mean substitution, Synthetic Minority Over-sampling Technique (SMOTE), and feature importance selection, followed by Logistic Regression and K-Nearest Neighbor classifiers. It has been proven to be highly accurate and efficient in databases such as Cleveland, heart disease, and Framingham.

Yaqoob et al.[21] proposed a Federated learning approach to predict cardiovascular diseases while preserving privacy. At client sites, they optimized their workflow for feature selection using Modified Artificial Bee Colony (MABC) optimization, classified using Radial Basis Support Vector Machine (RB-SVM) classifiers, and aggregated the results on their server using Federated Matched Averaging (FedMA). This design is extremely accurate, reduces communication, and maintains patient data privacy according to worldwide regulations. Rustam et al.[22] proposed a hybrid model that extracts enriched features from EHRs using CNN and classifies the features using a soft voting ensemble of logistic regression, SVM, and SGC. It was evaluated using four cardiovascular datasets: the heart failure prediction dataset, coronary heart disease dataset, structural heart disease dataset, and SAHDD, and was shown to enhance predictive accuracy and generalizability.

Balakrishnan et al.[23] developed an IoT based diagnostic system by using echocardiogram imaging, wearable sensors and fuzzy clustering for segmentation. They used a Pretrained Recurrent Convolutional Neural Network (PRCNN) and a Deep Convolutional Neural Network (DCNN) for risk stratification, and Wiener filtering was used for image preprocessing. The method resulted in close-to-perfect classification performance, which was better than that of traditional machine learning models. Narasimhan et al.[24] conducted empirical evaluations across datasets that included Cleveland, Faisalabad, and Framingham datasets. They employed different classifiers, such as random forest, KNN, SVM, logistic regression, gradient boost, naïve Bayes, and neural networks, using SelectKBest for feature selection, and showed the effect of the choice of dataset on the performance of the classifiers.

Shrivastava et al.[25] proposed a hybrid model of Convolutional Neural Networks (CNN) and Bidirectional Long Short-Term Memory (Bi-LSTM) named HCBiLSTM. The model fitted to the Cleveland dataset showed better performance in terms of accuracy, precision, recall, and F1-score than the standard classifiers after the features were selected using an Extra Trees Classifier based on Gini importance. Uddin et al.[26] carried out a wide comparison of ten machine learning algorithms (Decision Tree, SVM, Random Forest, XGBoost, AdaBoost, Gradient Boosting, Multilayer Perceptron, etc.) on the datasets collected from UCI, IEEE DataPort, and Kaggle. The results obtained by them indicated that the Decision Tree could classify the data with the highest accuracy. They also provided an online interface for clinical deployment and graphical analysis based on heatmaps and ROCs.

Ozcan et al.[27] obtained an interpretable model by combining five heart datasets (Cleveland, Hungarian, and Statlog) into an interpretable Classification and Regression Tree (CART) classifier. After cleaning, 745 records were analyzed, attributes were identified, and 18 decision rules were extracted to make the model transparent and aligned with clinical use cases. Majumder et al.[28] introduced the combination of any two base classifiers (SVM, Decision Tree, and KNN) and another combination (Logistic Regression, AdaBoost, and Naïve Bayes) by taking the final decision using voting. Feature selection was performed using Extra Trees, and normalization was performed using StandardScaler, which is the normalization technique used to increase the interpretability and generalization of the UCI Heart Disease dataset.

Gupta et al.[29] focused on feature selection using Genetic Algorithm (GA) and Cuckoo Search Optimization (CSO) and then used an ensemble of an aggregation of Random Forest and CNN based classifiers to classify the images using majority voting. The GA also tuned the hyperparameters,where preprocessing included outlier detection using the IQR and Min–Max normalization. They clearly demonstrated performance gains over the UCI. Doppala et al.[30] reduced the number of features from 14 to 9 using Genetic Algorithm and classified the features using a Radial Basis Function neural network. Their approach achieved a 94.2% accuracy compared to 85.4% and showed better results in terms of accuracy, precision, recall, F1-score, specificity, and Matthews Correlation Coefficient, making evolutionary feature reduction a powerful tool. A comparison of the existing techniques is presented in Table 1.

The systematic evaluation of the proposed framework was conducted using a Cardiovascular Disease Prediction Model Using Binary Krill Herd Algorithm (B-KHA) and Capsule Network (CapNet) by Saran et al.[12]. The results of the work revealed that the proposed framework performed better than other models due to enhanced prediction accuracy. To achieve superior results, Saran et al.[8] designed a Stacked Ensemble Framework Using Neural Network Reasoning and Hawks Optimizer (HO) for cardiovascular disease prediction. Moreover, the researchers addressed class imbalance in order to make the predictions reliable.

Arunachalam et al.[31] created a novel Ensemble-Based Cardiovascular Disease Prediction Framework that used K-Nearest Neighbor (KNN), eXtreme Gradient Boosting (XGBoost), AdaBoost, and Random Subspace

References	Methodology	Advantages	Limitations
Tiwari et al.[2]	Stacked ensemble combining top ML classifiers (e.g., XGBoost, ExtraTrees, Random Forest) selected via tenfold CV on a merged CVD dataset (1190 cases, 11 features)	Higher accuracy than individual models; leverages model diversity for better generalization on routine clinical features	Evaluated only on a single merged dataset with classical ML; deep learning/larger datasets left for future work
Khan et al.[3]	PCG acquisition with spectral feature extraction + ANN classifiers	High accuracy (99.99%), low-cost, portable, non-invasive system	Limited to binary classification (normal vs abnormal murmurs); lacks multi-class insight
Theerthagiri et al.[4]	RFE-guided feature selection + Gradient Boosting classifier on CVD tabular data	Outperforms LR, RF, MLP, SVM, NB in accuracy/AUC with compact feature set	Segmentation metrics (Dice/IoU/HD) and specificity not reported
Masumoto et al.[5]	Retrospective cohort analysis from CLAVIS-COVID database. Patients categorized by LDH tertiles (T1–T3) to assess mortality outcomes	Uses a large, multicenter cohort focusing specifically on cardiovascular comorbidities and LDH	Retrospective design, early pandemic data (2020), no post-discharge outcomes, LDH isoforms not analyzed
Sapra et al.[6]	Dual-phase model combining Deep Neural Network (with Gaussian noise) and Case-Based Reasoning	Achieves high diagnostic accuracy and severity detection using routine clinical data	Does not provide certain image-based or IoU/Dice metrics
Li et al.[7]	Multi-modal deep learning with ECG + PCG features, GA, SVM	Improved AUC and classification accuracy over single-modals	PCG network performance was limited due to dataset noise
Rekha et al.[8]	Stacked Ensemble + Sample-based NN Reasoning + Improved Hawks Optimizer (with chaotic maps and non-linear controls)	High accuracy (97%), improved convergence, balances exploration & exploitation, robust to imbalance using SMOTE	Complexity in algorithm due to integration of nonlinear and chaotic processes
Sumwiza et al.[9]	Developed an RF model using feature importance & correlation for heart disease prediction	Achieved 99% accuracy after preprocessing and outlier removal	Specificity and other metrics not reported
Kiran et al.[10]	GBDT + Binary Spotted Hyena Optimizer (BSHO)	High accuracy (97.89%), effective feature selection, robust performance across datasets	Specificity and real-world deployment not evaluated
Gupta et al.[11]	Stacking ensemble with deep neural networks and LogitBoost meta-learner trained via tenfold cross-validation on bootstrapped data	High accuracy and robustness for predicting heart disease in post-COVID patients	Limited dataset size, lacks external validation
Kumar et al.[12]	CapNet + B-KHA + Gaussian Optimizer + VGG-16 on CVD data	High accuracy, robust feature selection, less overfitting	Time complexity not yet evaluated
Khan et al.[13]	Two-stage PCG-based classification using EMD preprocessing, MFCC/statistical/spectral feature fusion, and KNN/SVM classifiers	Non-invasive, low-cost, suitable for embedded systems, detects CAD types	Precision/F1 not discussed, small data for some classes
Nelson et al.[14]	ABFog (AlexNet + Fog/Edge + IoT sensors + FogBus)	Real-time heart disease prediction with low latency	No real-time deployment tested; simulation only
Hasan et al.[15]	Multi-stage feature selection + XGBoost	High accuracy via ensemble-based robust feature filtering	Conducted on single, low-dimensional dataset
Baghdadi et al.[16]	Proposed CatBoost with Shapley value-based feature selection and tuning	High predictive performance with fewer features; clinically applicable and cost-efficient	Relies on secondary cross-sectional data; lacks longitudinal validation
Higaki et al.[17]	Retrospective cohort work of 601 hospitalized COVID-19 patients with CVDRF; used NLR at baseline and follow-up; applied t-SNE clustering	Shows dynamic importance of NLR over time; applied machine learning (t-SNE) for phenotyping	Baseline NLR not independently predictive; no control group without CVDRF
Reddy et al.[18]	Used Decision Tree and Gaussian Naïve Bayes on Heart Failure Dataset with pre-processing and feature selection	GNB showed better accuracy on real medical data	Limited dataset size; assumptions of NB may not always hold
Sinha et al.[19]	Hybrid ML model combining 4 feature selection methods (RFE, LASSO, Chi-Square, Tree-Based) and 7 classifiers including SVM	High prediction accuracy (92.31%) with optimized feature sets. Integrated with IoMT for real-time use	Requires multiple model training (112 total); no specificity values reported
Rahim et al.[20]	Proposed MaLCaDD framework integrating preprocessing (outlier removal, mean substitution, SMOTE), feature selection, and ensemble (LR + KNN)	Achieved very high accuracy (up to 99.1%) with reduced features, less computational complexity	Other evaluation metrics (precision, recall, F1) not reported; real-world hospital data testing still in progress
Yaqoob et al.[21]	MABC + RB-SVM at clients with FedMA on server	Improved privacy, accuracy, communication cost	No comparison with deep learning techniques
Rustam et al.[22]	CNN-based feature expansion + Ensemble ML (LR, SVM, SGDC)	Boosted accuracy using CNN-enhanced features	Increased computational complexity; limited generalization on small/imbalanced datasets
Balakrishnan et al.[23]	IoT-based echocardiogram classification using FCM + PRCNN + DCNN	High accuracy (99.5%) in classification and risk prediction	No mention of validation on external datasets
Narasimhan et al.[24]	Two-phase approach: (1) Feature selection using SelectKBest; (2) Classification using ML models (RF, KNN, SVM, NB, LR, XGBoost, NN)	Random Forest achieved the highest accuracy and lowest classification errors across datasets	Limited metrics reported (only accuracy and error); lacks comprehensive evaluation (precision, recall, F1, specificity)
Shrivastava et al.[25]	Hybrid CNN-BiLSTM with Extra Tree Classifier for feature selection	High accuracy in heart disease prediction	Specificity not evaluated
Uddin et al.[26]	Used 10 ML models on a combined heart disease dataset; DT model deployed via web app	Achieved 99.16% accuracy using DT; created a public web interface for prediction	Limited discussion on explainability; results not validated externally
Ozcan et al.[27]	Used CART decision tree on curated dataset of 745 records for heart disease prediction with rule extraction	High interpretability via decision rules; good accuracy; practical clinical applicability	Limited feature diversity (e.g., social factors excluded); unstructured data not used
Majumder et al.[28]	Concatenated hybrid ensemble classifier combining two hybrid models	Improved accuracy, robustness, and interpretability	Not tested on larger or diverse clinical datasets
Gupta et al.[29]	Hybrid feature selection (GA + CSO) + Ensemble learning (RF + CNN + Majority Voting)	High accuracy, robust performance, handles feature selection	Limited dataset size (303 instances in UCI)
Doppala et al.[30]	GA + RBF (Genetic Algorithm + Radial Basis Function)	High accuracy after feature selection; reduced number of attributes; computational efficiency	Evaluation limited to small dataset (297 records); needs testing on larger datasets

Table 1. Comparative analysis of the existing techniques.

classifiers. In the research, the proposed model proved more accurate because of efficient feature selection by using Linear Support Vector Feature Measure (LSVFM). Another interesting framework for cardiovascular disease prediction has been designed by Saranya et al.[32]. It involves the use of a hybrid deep learning (HDL) model called DenseNet-Attention-based BiLSTM to classify multiclass arrhythmias. Finally, Rajagopal et al.[33] suggested using dimensionality reduction and PNN classifier for arrhythmia classification.

According to Solanki et al.[34] a Convolutional Neural Network (CNN) framework employing Visual Geometry Group-16 (VGG16) and Visual Geometry Group-19 (VGG19) models was utilized to classify brain tumors. The VGG19 model yielded superior classification accuracy compared to other deep learning models. In the work by Selot et al., a comparison between Convolutional Neural Networks (CNN) and Support Vector Machines (SVM) for Alzheimer's disease prediction was presented[35]. According to experimental results, the SVM model obtained superior predictive accuracy compared to the CNN model. In another work by Siddiqui et al., a novel CBAMWDnet deep learning model for tuberculosis classification based on chest X-ray (CXR) images was designed[36].

Wang et al.[37] have introduced a framework of joint learning by incorporating the concepts of COVID-Net and contrastive learning for the classification of COVID-19 cases through Computed Tomography (CT) images. This framework has enhanced the accuracy of the classification process for heterogeneous data sets from various sites. Xu et al.[38] introduced an approach of domain adaptation for predicting the severity of COVID-19 using Computed Tomography (CT) images. In order to handle class imbalance and domain mismatch issues, this framework improves the accuracy of severity classification process. Yang et al.[39] have introduced transfer learning and site-specific approaches for COVID-19 classification.

Cardiovascular complications in people who have recovered from COVID-19 have added new challenges to diagnosis that are not addressed by traditional tools and predictive models. While classical methods for the detection of CVDs rely on pre-established diagnostic pathways, SARS-COV-2 (COVID-19) affects the body in a systemic manner, leading to unique changes in biomolecules and functional systems. This limits a clinician's ability to predict cardiac events in patients without a history of heart disease who are showing up as vulnerable due to a past infection. Moreover, there are no intelligent, scalable systems that can integrate clinical, biochemical, and physiological variables reflecting post-viral changes. Current AI models either depend on historical data or do not adapt to changing heart profiles. This creates an urgent need for a robust, data-driven solution that can dynamically evaluate cardiac risk in the post-pandemic context—accounting for both traditional risk factors and the newly emerging indicators associated with Post-COVID-19 patients and related complications.

Methodology

The innovation of the proposed framework does not only lie in the incorporation of various processing modules, but rather in the creation of a unique multimodal DIR-GAN based learning approach that incorporates both imaging modalities and structured clinical information to predict cardiovascular risks in patients who have survived from COVID-19. In contrast to typical multimodal deep learning techniques that aggregate features directly before performing deterministic classification, the proposed technique involves adversarial multimodal learning whereby generated fusions of imaging and clinical data are optimized through the process of generator and discriminator competition, making the model more robust to heterogeneities and noise present in post-COVID physiological conditions. Additionally, the fusion of SSECT-based myocardial texture analysis and ASO feature optimization creates more discriminative and succinct feature representation before the learning task begins. Thus, the contribution of the proposed approach can be said to be its combination of multimodal feature fusion, feature optimization, and adversarial discriminative learning in the prediction of post-COVID cardiac risk assessment.

Overall, this pipeline—spanning ABF-based enhancement, KDFCM segmentation, SSECT feature extraction, clinical data refinement via Isolation Forest and ASO, feature-level fusion, and final DIR-GAN classification—provides a systematic and reliable method for forecasting cardiovascular outcomes in Post-COVID-19 patients as shown in Fig. 1.

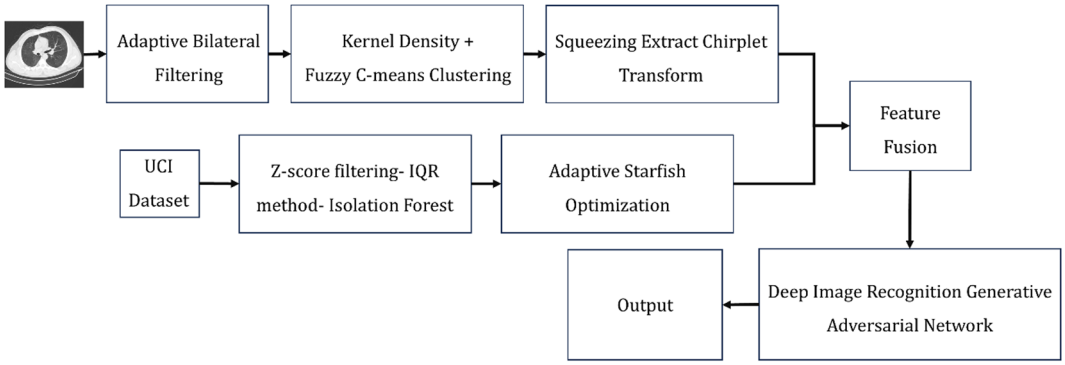


Fig. 1. Overall flow diagram of the proposed framework.

Data acquisition

For this work, the dataset comprised anonymized CT scanning data and structured cardiovascular data from the UCI repository. Complementarily, these two modalities provided both visual and clinical views, which together served as a solid basis for the proposed predictive framework.

Imaging data

The imaging part of this work comprised a dataset of 746 computed tomography (CT) scan images, which were split into two diagnostic sets. Of these, 349 scans were from patients who were positive for COVID-19 and 397 scans were from patients who were negative for COVID-19. This even distribution resulted in meaningful comparisons between groups and made it possible for the system to grasp the structural differences in the cardiac region related to infection. Both scans were obtained at diagnostic quality and had a high enough resolution to preserve myocardial detail for subsequent enhancement and segmentation. All scans were anonymized prior to analysis to prevent patient identification. Images with severe motion artifacts and missing clinical annotations were excluded to ensure the reliability of the images.

Even though the structured clinical data was retrieved from a classical cardiovascular disease database and not from post-COVID patients, its use was justified by the known connection between traditional cardiovascular risk factors and post-COVID cardiovascular disorders. It has been previously observed in clinical studies that post-COVID cardiovascular symptoms tend to be linked with signs that suggest pre-existing cardiac susceptibility such as blood pressure deviations, altered cholesterol levels, impaired physical endurance, abnormal ECG readings, and impaired myocardial function. For this reason, the structured clinical data from the UCI heart disease database was used for providing biologically relevant cardiovascular markers, whereas CT images allowed extracting structural features related to physiological changes due to COVID. In other words, the designed multimodal approach can be considered as a cardiac vulnerability prediction model and not as a clinically tested post-COVID outcome predictor. The approach was chosen due to the availability of publicly available multimodal data and physiological relevance to cardiovascular risk prediction.

Clinical data

In addition to imaging, a cardiovascular dataset obtained from the UCI Machine Learning Repository was employed to provide structured clinical records. This dataset included patient-level information across multiple demographic, diagnostic, and physiological features. The attributes are summarized as follows:

- Age (29–77 years) and Sex (male = 1, female = 0).
- Chest pain type (CP) classified into four categories: typical angina, atypical angina, non-typical angina, and asymptomatic (coded as CP = 1–4).
- Resting blood pressure (RestBP) ranging between 94 and 200 mmHg.
- Serum cholesterol (Chol) levels spanning 126–564 mg/dl.
- Fasting blood sugar (FBS) recorded as binary values depending on whether glucose exceeded 120 mg/dl (1 = yes, 0 = no).
- Resting ECG results graded 0–2, reflecting varying degrees of abnormality.
- Maximum heart rate achieved (Heart Beat) ranging from 71 to 202 bpm.
- Exercise-induced angina (Exang) noted as 1 = yes, 0 = no.
- Oldpeak, an index of ST depression, varying from 0.0 to 6.2.
- Slope of ST segment during exercise stress test: upsloping (1), flat (2), or downsloping (3).
- Number of major vessels (CA) visible via fluoroscopy, ranging from 0 to 3.
- Thalassemia status (Thal) encoded from 0 to 3.
- Target variable indicating the presence (1) or absence (0) of cardiovascular disease.

This structured dataset contained several patient records, but only a small number had missing or inconsistent entries. Only records that would maintain the statistical integrity of the study were included in the preprocessing stage. All imaging and clinical data were acquired from public data repositories with anonymized patient information. Ethical approval was not required as it was clear that all identifying details had been removed before the distribution. However, inclusion/exclusion criteria were also strictly applied; only complete cases with good-quality scans and all attributes filled in were included, while corrupted, incomplete, or low-quality cases were removed. This ensured that the analysis was conducted on reliable and ethically compliant data.

While the framework incorporates imaging data and clinical data that are freely available in order to enhance reproducibility, there are some limitations that need to be addressed. First, the clinical data set was collected from the UCI Heart Disease Data Set, which mostly includes traditional factors of heart disease risk and may not provide the best results for identifying changes in physiology related to cardiovascular issues in post-COVID patients. Second, the imaging dataset included 746 images; however, demographics, such as the age of the participants, the number of male and female participants, and their ethnic backgrounds and comorbidities were missing from the repository. As a result, the demographic characteristics of the imaging dataset could not be assessed properly.

As a result of the independence of the image database and the structured clinical database, with neither providing any subject-wise matching, direct matching between the two at the patient level was not possible. This led to the multimodal fusion process being performed at the level of feature representations rather than on a per-patient basis. In this case, myocardial image features that are linked to changes in structure after contracting COVID-19 are extracted using cardiac CT scans, while physiological cardiovascular risk-related features are taken from clinical data from the UCI repository. This resulted in feature-based multimodal fusion using the preprocessed and normalized representative image and clinical features.

However, the suggested approach is not based on labeled information about post-COVID outcomes, as such annotations were not accessible in the publicly available data used in the present work. Instead, the approach was based on combining multimodal data about COVID-19 CT patterns and conventional cardiovascular biomarkers. In more detail, the CT data allowed obtaining features reflecting myocardial changes due to physiologic changes induced by the viral infection. The other input dataset (UCI cardiovascular one) included well-known cardiovascular risk markers. After performing the necessary steps of data preprocessing, feature extraction, and multimodal fusion, the DIR-GAN framework discovered discriminatory relations between the myocardial abnormalities obtained using imaging and cardiovascular biomarkers and predicted cardiovascular risk in a post-COVID setting. Thus, the suggested approach can be treated as a predictive tool for cardiovascular risk rather than post-COVID outcome prediction.

Data augmentation

Various data augmentation methods were used for augmenting the cardiac CT scans in the preprocessing phase. Rotation, flip, scale, and some other slight variations in the intensity level of images were done while keeping the medically important characteristics of the myocardium intact. These methods of data augmentation were chosen because they can provide variations in the imaging process and orientation of images without altering any pathological features of post-COVID heart disorders.

Preprocessing

Since the proposed methodology uses heterogeneous datasets, preprocessing was tailored according to the variety of imaging and clinical attributes. The CT scan image data is characterized by different resolutions, contrasts, scanning techniques, and levels of noise; hence, Adaptive Bilateral Filtering (ABF) was used to reduce the noise while retaining the edges of the myocardium for reliable segmentation and feature extraction. On the other hand, structured clinical attributes are significantly different in scale, distribution, and physiological interpretation; thus, Z-score standardization, interquartile range-based outlier elimination, and isolation forest-based anomaly detection were used.

Imaging preprocessing

Noise is caused by acquisition hardware, patient movements and environmental noise in magnetic resonance and computed tomography images. For this purpose, an Adaptive Bilateral Filter (ABF) was used for every image. This method preserves the sharpness of significant edges while attenuating unwanted variations in homogeneous areas, unlike simple smoothing operations. The filter operates by computing the geometric proximity of the neighbors and the similarity of the gray levels of the neighbors. This relation could be mathematically represented as

$$I_{filtered}(p) = \frac{1}{W_p} \sum_{q \in \Omega} I(q) \cdot \exp\left(-\frac{\|p-q\|^2}{2\sigma_s^2}\right) \cdot \exp\left(-\frac{\|I(p)-I(q)\|^2}{2\sigma_r^2}\right) \quad (1)$$

where I(p) is the intensity of the pixel under consideration, q represents a neighboring pixel within the window Ω, σ_s controls the influence of spatial distance, and σ_r regulates the sensitivity to intensity differences. The normalization term W_p ensures that the output remains within valid image intensity bounds. In its adaptive form, the filter dynamically adjusts the range parameter σ_r according to local contrast, making it particularly effective for myocardial analysis. This step is essential because clear visualization of ventricular boundaries and myocardial walls is critical for accurate segmentation, which in turn influences the precision of texture and frequency feature extraction.

Clinical data preprocessing

The clinical dataset contains variables that differ widely in their ranges and distributions, making it unsuitable for direct integration with imaging-derived features. To achieve uniformity, all variables are transformed using Z-score normalization. Each value is rescaled according to

$$z_i = \frac{x_i - \mu}{\sigma} \quad (2)$$

where x_i is the original value, μ is the mean of the feature, and σ is the standard deviation. After this transformation, every variable has zero mean and unit variance, ensuring that no single measurement dominates the learning process due to differences in scale.

Even after normalization, certain observations may lie far outside the general distribution and risk biasing the analysis. To remove these, the interquartile range (IQR) approach is applied. This measure is calculated as

$$IQR = Q_3 - Q_1 \quad (3)$$

where Q_1 and Q_3 denote the first and third quartiles. Any observation that falls below $Q_1 - 1.5 \times IQR$ or exceeds $Q_3 + 1.5 \times IQR$ is flagged as an outlier and excluded from the dataset. By using this non-parametric approach, the filtering process remains effective even when data distributions deviate from normality.

To further refine the dataset and detect more subtle irregularities, an Isolation Forest algorithm is employed. Unlike density-based or distance-based approaches, this method isolates anomalies through recursive partitioning. Each record is assigned an anomaly score derived from the average path length E(h(x)) of the observation x across a collection of random decision trees:

$$s(x,n) = 2^{-\frac{E(h(x))}{c(n)}} \tag{4}$$

The shorter the path length, the more easily the observation can be isolated; hence, the more likely the observation is to represent an anomaly. This is especially effective for high-dimensional clinical data, where it may be possible to identify abnormal records that would not be identified by simple statistical thresholds.

Normalization, outlier exclusion, and anomaly detection provide a combined effect that guarantees that the clinical dataset is consistent, balanced, and free of distortions. In combination with the denoised imaging data, this preprocessing approach ensures that the subsequent feature extraction and fusion processes are conducted on well-structured and representative patient information, thus enhancing the robustness of the predictive framework.

Imaging feature extraction

To accurately segment the myocardium, the imaging feature extraction stage guarantees that the noisy cardiac images are first accurately segmented by KDFCM, which preserves the structural boundaries. Next, SSECT is used to obtain multi-scale texture and frequency patterns, which can help detect subtle changes in tissues that are related to cardiovascular disease.

Segmentation: kernel density estimation with fuzzy C-means (KDFCM)

Good delineation of the myocardium is essential for obtaining meaningful cardiac biomarkers from medical images. However, conventional clustering techniques are generally not successful when overlapping intensity distributions are present, as is frequently the case in cardiac CT and MRI because of partial volume effects and noise. To overcome this challenge, myocardial segmentation was performed using Kernel Density Estimation with Fuzzy C-Means (KDFCM).

The fuzzy C-means (FCM) clustering algorithm assigns each pixel x_j to clusters with varying degrees of membership, minimizing the objective function:

$$J = \sum_{i=1}^{c} \sum_{j=1}^{N} u_{ij}^{m} \left\| x_j - v_i \right\|^2 \tag{5}$$

where c is the number of clusters, N is the total number of pixels, u_{ij} represents the membership degree of pixel x_j in cluster i, v_i is the centroid of cluster i, and $m > 1$ is the fuzzification parameter that controls cluster overlap.

To enhance segmentation robustness, kernel density estimation (KDE) was integrated with FCM. KDE estimates the probability density function of the image data as:

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right) \tag{6}$$

where K is the kernel function, h is the bandwidth parameter, and x_i are the sample points. By embedding KDE into the clustering process, KDFCM better separates regions with subtle intensity variations, ensuring precise delineation of the myocardium. This segmentation step isolates the cardiac tissue of interest while excluding surrounding structures, providing a reliable basis for feature extraction.

Feature extraction: squeezing extract chirplet transform (SSECT)

Once the myocardium was segmented, the next step was to capture its textural and frequency-based characteristics using the Squeezing Extract Chirplet Transform (SSECT). Traditional Fourier-based methods often lose time-localized information, while wavelet transforms may not effectively represent oscillatory signals with varying frequency modulations. SSECT overcomes these limitations by decomposing the myocardial image into chirplet basis functions, which are capable of representing both localized textures and frequency sweeps across multiple scales.

For the detection of abnormalities related to the myocardium caused by post-COVID myocarditis, the parameters for the chirplet basis function model were tuned based on the patterns observed in physiological changes induced due to the inflammatory and fibrotic state of the myocardium, as well as myocardial tissue heterogeneity and microvascular injuries. In the case of Eq. (7), the t_0 parameter allows for the identification of local regions of change in myocardium conditions. Additionally, σ ensures that myocardial textures remain continuous even when localized without excessively smoothing away myocardial borders. The f_0parameter represents the dominant oscillations in myocardial textures caused by inflammatory or ischemic changes. Lastly, the chirp-rate parameter α is used for modeling nonstationary frequencies, which represent myocardial deformations.

A chirplet atom is defined as:

$$g(t) = \exp\left(-\pi \frac{(t - t_0)^2}{\sigma^2}\right) \cdot \exp\left(j\left(2\pi f_0 t + \alpha t^2\right)\right) \tag{7}$$

where t_0 is the time shift, σ controls time spread, f_0 is the central frequency, and α is the chirp rate. By adjusting these parameters, chirplets can model localized oscillations with varying frequency trends.

The chirplet atom is described in Eq. (7), which represents a local basis that models the time–frequency behavior of myocardial dynamics. Yet, when obtaining the discriminating feature representations, these chirplet atoms are mapped onto the myocardial signal via the SSECT process shown in Eq. (8). By so doing, spectral energy can be more concentrated in the time–frequency space, making it more sensitive to frequency changes and heterogeneous textures resulting from post-COVID myocardial remodeling.

The SSECT applies a squeezing operator to reassign time–frequency energy into sharper distributions, defined as:

$$T_{SSECT}(t,f) = \int s(\tau)\,\psi^{*}_{t,f}(\tau)\,d\tau \tag{8}$$

where s(τ) is the input myocardial signal, and $\psi_{t,f}$ denotes the chirplet basis centered at time t and frequency f. This results in high-resolution multi-scale descriptors that encode myocardial textures, subtle fibrosis patterns, and frequency irregularities linked with cardiovascular dysfunction.

The combined use of KDFCM segmentation and SSECT feature extraction ensures that the imaging branch captures both structural boundaries and functional textural variations of the myocardium. Segmentation guarantees that only relevant cardiac tissue contributes to analysis, while SSECT extracts discriminative biomarkers such as irregular frequency components associated with ischemia or abnormal textures suggestive of scarring. Together, these features provide a robust imaging-based signature of cardiovascular disease risk in Post-COVID-19 patients.

After undergoing transformation through SSECT, various statistical properties were derived from the transformed myocardial feature representation to identify differences in texture and intensity due to the presence of cardiovascular anomalies. The first-order statistical properties, which include mean, standard deviation, variance, skewness, kurtosis, entropy, and energy, were determined using the transformed myocardial feature representation to provide quantitative analysis of signal distribution, variability, structural irregularity, and local heterogeneity. By extracting statistical information on texture and frequency variation from the transformed feature representation, it is possible to quantitatively determine cardiovascular abnormalities caused by COVID-19 infection. The derived statistical information was used as the input for multimodal feature fusion and cardiovascular risk assessment.

Clinical data feature engineering

Following the preprocessing stage, the clinical dataset required refinement to reduce redundancy and retain only the most informative features for cardiovascular risk prediction. High-dimensional data often contains irrelevant or correlated attributes that can reduce classification accuracy and increase computational complexity. To address this challenge, a two-step approach was employed: feature selection and optimization.

Feature selection and dimensionality reduction

The first step involved evaluating the contribution of each feature toward predicting cardiovascular outcomes. Let the dataset be represented as:

$$D = \{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\} \tag{9}$$

where $x_i \in \mathbb{R}^d$ is the feature vector of dimension d, and y_i is the target label indicating disease presence or absence. The goal of feature selection is to identify a reduced subset $F \subseteq \{1,2,\ldots,d\}$ such that the selected features maximize predictive performance while minimizing redundancy.

Dimensionality reduction was achieved by ranking features based on statistical significance and correlation analysis, followed by the application of an optimization algorithm to identify the most discriminative combination.

Adaptive Starfish Optimization (ASO)

To refine the feature subset, the Adaptive Starfish Optimization (ASO) algorithm was applied. Inspired by the foraging, regeneration, and adaptive movement behavior of starfish, ASO balances global exploration and local exploitation as shown in Fig. 2. Each candidate solution represents a potential subset of features, encoded as a binary vector:

$$S = (s_1, s_2, \ldots, s_d),\ s_j \in \{0,1\} \tag{10}$$

where $s_j = 1$ indicates selection of the j-th feature, and $s_j = 0$ indicates exclusion.

The fitness function was designed to jointly evaluate classification accuracy and the number of selected features:

$$f(S) = \alpha \cdot Accuracy(S) - \beta \cdot \frac{|S|}{d} \tag{11}$$

where α and β are weighting factors that balance predictive power and dimensionality.

During each iteration, ASO updates candidate solutions through three key operators:

- Radial Movement – simulates starfish extending arms to explore new regions in the search space.

$$S^{(t+1)} = S^{t} + r \cdot (S_{best} - S^{t}) \tag{12}$$

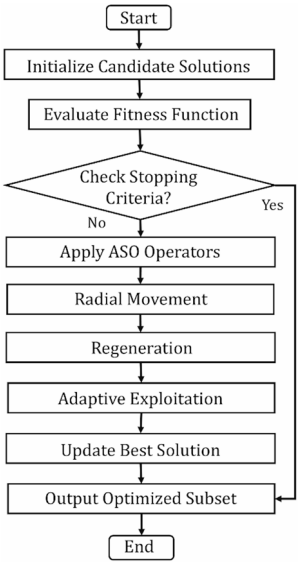


Fig. 2. Flowchart of the Adaptive Starfish Optimization (ASO) algorithm for clinical feature selection.

where r is a random factor and S_{best} is the best-known solution.

- Regeneration – mimics the ability of starfish to regrow arms, allowing the algorithm to recover lost diversity by reintroducing random features.
- Adaptive Exploitation – fine-tunes promising solutions by adjusting feature selection probabilities based on feedback from the fitness function.

The iterative process continues until convergence, yielding an optimized subset of clinical biomarkers that carry the strongest predictive power for cardiovascular disease.

By applying ASO, features such as age, cholesterol level, maximum heart rate, and ST-depression (Oldpeak) emerged as highly influential clinical indicators. These biomarkers, when combined with imaging features, provide a comprehensive view of patient-specific cardiovascular risk, enabling early detection and more personalized management strategies.

Feature-level fusion

To enhance the performance of predicting cardiovascular risk, imaging features from cardiac CT images and clinical biomarkers from structured patient medical records were combined using feature-level fusion techniques. In this way, data from both sources could be combined effectively and used for the purpose of classification, which would allow the system to leverage the relationship between myocardial structure and cardiovascular health.

Let the imaging feature vector be:

$$F_{img} = \{f_1, f_2, \ldots, f_m\} \in \mathbb{R}^m \tag{13}$$

and the clinical feature vector be:

$$F_{clin} = \{c_1, c_2, \ldots, c_n\} \in \mathbb{R}^n \tag{14}$$

where m and n denote the feature dimensions of imaging and clinical modalities, respectively.

The multimodal feature representation was constructed through feature concatenation:

$$F_{fusion} = [F_{img} \parallel F_{clin}] \tag{15}$$

where $\parallel$ denotes vector concatenation, generating a composite feature vector of dimension $m + n$.

To ensure balanced contribution from both modalities and avoid dominance of high-dimensional imaging features, modality-specific weighting coefficients were applied:

$$F_{fusion} = \alpha F_{img} \parallel \beta F_{clin} \tag{16}$$

subject to: $\alpha + \beta = 1$, where α and β represent weighting factors for imaging and clinical features, respectively. The fused representation was subsequently used as input to the DIR-GAN framework for cardiovascular risk prediction.

Classification

The last step in the proposed pipeline is the classification of patients into cardiovascular disease (CVD) risk categories using a deep image recognition-based Generative Adversarial Network (DIR-GAN). This framework is based on an adversarial learning approach that not only boosts the discriminative power of the model but also increases the robustness of the features, in contrast to traditional classifiers that simply map features to labels. The architecture includes a generative part as a component of the classification process, which decreases the risk of overfitting to the training data and helps generalize well with respect to unseen patient cohorts.

At the core of the DIR-GAN lies the interaction between two neural networks, the generator and the discriminator. The generator, denoted as G, is trained to transform latent feature vectors z into synthetic feature representations G(z) that closely resemble the fused multimodal features extracted from patient data. The discriminator, denoted as D, is simultaneously trained to distinguish between real patient-derived features and the synthetic ones produced by the generator. The adversarial training objective can be expressed as:

$$\min_G \max_D V(D,G) = \mathbb{E}_{x\sim p_{data}(x)}[\log D(x)] + \mathbb{E}_{z\sim p_z(z)}[\log(1 - D(G(z)))] \tag{17}$$

Here, $p_{data}(x)$ represents the true distribution of multimodal patient features, while $p_z(z)$ denotes the prior distribution used to sample latent variables. The discriminator attempts to maximize its ability to correctly classify true versus generated features, while the generator seeks to minimize the discriminator's success by producing increasingly realistic synthetic features.

In the proposed framework, the discriminator is extended beyond a binary "real or fake" decision. Instead, it is designed to function as a multi-class classifier that assigns patient feature vectors to specific CVD risk categories, such as low, moderate, or high risk as shown in Fig. 3. For an input feature vector x, the discriminator outputs class probabilities through a softmax function:

$$P(y = k|x) = \frac{\exp\left(W_k^T h(x) + b_k\right)}{\sum_{j=1}^{K} \exp\left(W_j^T h(x) + b_j\right)} \tag{18}$$

where h(x) represents the hidden representation learned by the discriminator, W_k and b_k denote the parameters associated with class k, and K is the number of risk categories. The cross-entropy loss is then used to optimize the classification performance:

$$\mathcal{L}_{cls} = -\sum_{i=1}^{N}\sum_{k=1}^{K} y_{ik} \log P(y = k|x_i) \tag{19}$$

where y_{ik} is the true label indicator for sample i and class k. This dual-role setup allows the discriminator to simultaneously distinguish synthetic from real data and to categorize patients into clinically meaningful outcomes.

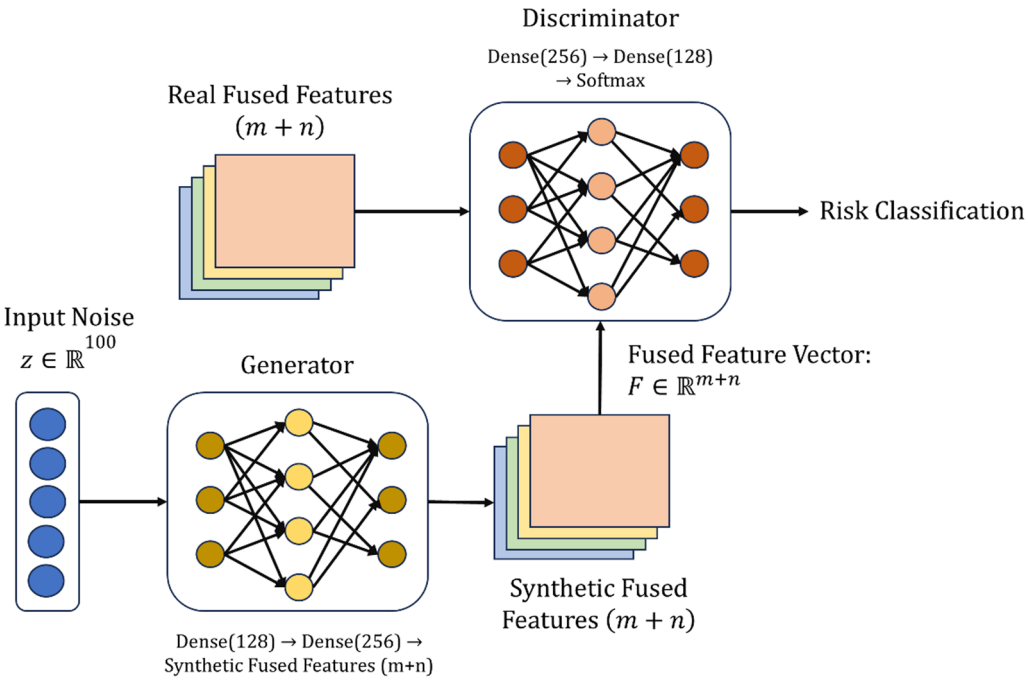


Fig. 3. Architecture of the DIR-GAN with generator and discriminator components.

The generator indirectly helps the classifier to be more robust to perturbed and synthetic data, which helps reinforce the decision boundary of the discriminator. Consequently, the discriminator can detect small differences in patient traits that are highly correlated with the risk of cardiovascular diseases. This competitive learning allows to counter the tendency of overfitting because the discriminator is not allowed to "learn" the training set by memorizing it, but must rather learn "rules" that can be applied to other data. Moreover, because the generator introduces additional variability into the feature space, the resulting model demonstrates improved resilience to noise and incomplete data, conditions that are common in real-world clinical scenarios.

The proposed DIR-GAN architecture comprises a generator and discriminator model that learn through an adversarial process to predict cardiovascular risk scores. The generator takes the multimodal features vector as the input and comprises three fully connected layers having 256, 128, and 64 nodes with ReLU activation function to create refined synthetic feature distributions from the compact feature representation learned by the model. On the other hand, the discriminator model consists of a multi-class classification network comprising three fully connected layers having 128, 64, and 32 nodes with the ReLU activation function along with a final Softmax layer for predicting low, moderate, and high risk of cardiovascular disease. Batch Normalization and dropout (rate of dropout = 0.3) techniques have been used for improving training process and preventing over-fitting. Training of model was done using Adam optimizer with a learning rate of 0.001, a batch size of 32, and categorical cross-entropy loss function. Adversarial training involved producing difficult examples by generator while discriminator learned to discriminate between real and generated feature representation.

Input: Cardiac CT image dataset I_{CT}, clinical dataset C_{data}
Output: Predicted cardiovascular risk category R

Begin
Acquire cardiac CT image dataset I_{CT} and clinical dataset C_{data}.
For each cardiac CT image $I_i \in I_{CT}$
Perform image enhancement using Adaptive Bilateral Filtering:
$I_{enh} = ABF(I_i)$
Perform myocardial segmentation using Kernel Density Fuzzy C-Means clustering:
$I_{myo} = KDFCM(I_{enh})$
Apply Squeezing Extract Chirplet Transform (SSECT) to extract imaging features:
$F_{SSECT} = SSECT(I_{myo})$
Extract statistical descriptors:
$F_{stat} = \{\mu, \sigma^2, \sigma, Sk, K, E, H\}$
where μ denotes mean, σ^2 variance, σ standard deviation,
Sk skewness, K kurtosis, E energy, and H entropy.
Generate imaging feature vector:
$F_{img} = F_{SSECT} \cup F_{stat}$
Normalize clinical dataset using Z-score normalization:
$C_{norm} = \frac{C_{data} - \mu}{\sigma}$
Remove outliers using Interquartile Range (IQR):
$C_{out} = IQR(C_{norm})$
Detect anomalous samples using Isolation Forest:
$C_{ref} = IsolationForest(C_{out})$
Select optimal biomarkers using Adaptive Starfish Optimization (ASO):
$F_{clin} = ASO(C_{ref})$
Perform multimodal feature fusion:
$F_{fused} = \alpha F_{img} + \beta F_{clin}$
where α and β denote weighting coefficients for imaging and clinical features.
Initialize Generator G and Discriminator D of DIR-GAN.
While convergence criterion is not satisfied
Generate synthetic fused features:
$F_{syn} = G(z)$
Train discriminator using real and synthetic features:
$D(F_{fused}, F_{syn})$
Update generator and discriminator parameters.
Predict cardiovascular risk category using trained DIR-GAN:
$R = DIR\text{-}GAN(F_{fused})$
Return cardiovascular risk category:
$R \in$ {Low Risk,Moderate Risk,High Risk}
End

Algorithm 1. Proposed DIR-GAN-Based Cardiovascular Risk Prediction Framework

Result and discussion

In this section, the Results and Discussion part gives a thorough analysis of the results obtained after performing experiments with the proposed DIR-GAN method. Experiments conducted on the dataset have been thoroughly evaluated and discussed with different evaluation measures such as accuracy, sensitivity, specificity, precision, and F1-score to understand the performance of cardiovascular risk prediction in patients with Post-COVID-19. The results obtained are compared against classical machine learning methods, deep learning, optimization algorithms, and current state-of-the-art techniques to demonstrate the effectiveness of the proposed methodology. In addition, the discussion part also analyzes the contribution of each component of the model, along with discussing computational feasibility and limitation of the model.

Experimental configuration

The problem of class imbalance was addressed so that model learning could be done effectively, and predictions made about the cardiovascular risks posed by different groups of patients accurately. This was achieved through

the use of balance sampling techniques and data augmentations in model training to overcome any problems associated with having a higher number of samples in dominant classes and to increase sample representation of minor classes. Moreover, multimodal feature fusion and adversarial learning in the DIR-GAN network helped achieve class discrimination by facilitating effective feature learning in diverse data types.

The proposed framework was developed and evaluated using a custom-built workstation configured with an Intel® Core™ i9-12900K processor clocked at 3.2 GHz, paired with 64 gigabytes of DDR5 memory to ensure fast data handling and multitasking. GPU-based acceleration was enabled via an NVIDIA® RTX 4090 graphics card featuring 24 GB of dedicated video memory, allowing efficient execution of deep learning tasks involving large-scale medical imaging data. All environments were implemented on Ubuntu 22.04 LTS for its ability to support CUDA-enabled operations and scientific computing frameworks. The code was developed in Python (v3.10), and the main deep learning library PyTorch (v2.0.1) was used to build and train the multimodal DIR-GAN-based classification model. OpenCV was used for image enhancement and image segmentation workflows, and NumPy and Pandas were used to handle structured clinical data and feature matrices. The hyperparameter values in Table 2 were obtained by gradually tuning the parameters over several training epochs based on the results of the parameter grid search and analysis of the convergence behavior. The final architecture was designed to provide a uniform performance for all the evaluation folds.

The evaluation technique was carefully crafted to ensure consistent and reliable predictions of the models on cardiovascular risk. The data were split into training and test sets in the ratio of 80:20, meaning that the model training was done based on the use of 80 percent of the samples, while the remaining 20 percent of the samples were used independently for model testing. Cross-validation with a fold size of 5 was adopted during the modeling process to enhance reliability. As the imaging data and clinical data were obtained from different public repositories, there were no matches between CT images and clinical data on an individual subject. Hence, multimodal integration was conducted based on extracted features after independent image preprocessing. In addition, parameter values for all models used were determined through experimental iterations aimed at obtaining stable and reliable results. The parameters for preprocessing and segmentation using CT images and KDFCM, respectively, were fine-tuned such that they preserved the myocardial structure homogeneity, whereas SSECT parameters were set based on empirical considerations so that they balanced out the texture and multiscale properties. Parameters for clinical data preprocessing, which included normalization and anomaly detection, among others, were established with consideration of the heterogeneity of the input features. In the same way, ASO features and DIR-GAN training parameters were tuned experimentally to achieve minimum classification errors.

Experimental analysis

The training and validation curves illustrate the strong learning capability and generalization power of the proposed model. The training and validation curves increased until a certain number of epochs, after which they plateaued at high values. The validation curve is almost parallel to the training curve, there is little overfitting, and the accuracy is consistent with the unseen data. At the last epoch, the model achieved an impressive accuracy of 94.98%, demonstrating its strength in classification tasks. Simultaneously, the training and validation losses continue to decrease steadily to low and stable levels, ending at 0.1953. The small difference between the two loss curves shows that the optimization is effective, the bias–variance trade-off is not too severe, and the convergence is reliable. The combination of these results demonstrates that the framework can learn complex patterns without compromising its generalization capability and can demonstrate predictive stability over different datasets, as shown in Fig. 4.

Figure 4 shows the training and validation curves for accuracy and loss with increasing epochs. The confusion matrix was used to showcase the accuracy of the proposed model in differentiating patients with and without cardiovascular disease. The results demonstrated that the disease was well detected, with 93.3% of actual positive

Hyperparameter	Value
Learning rate	0.0002
Batch size	32
Optimizer	Adam
Beta1 (Adam)	0.5
Beta2 (Adam)	0.999
Epochs	150
Dropout rate (Discriminator)	0.3
Dropout rate (Generator)	0.25
Latent vector dimension (z)	100
Generator hidden layers	3
Discriminator hidden layers	4
Feature fusion weight (α:β)	0.6:0.4
SSECT window size	64 × 64
Number of SSECT scales	5

Table 2. Proposed hyperparameter and their values.

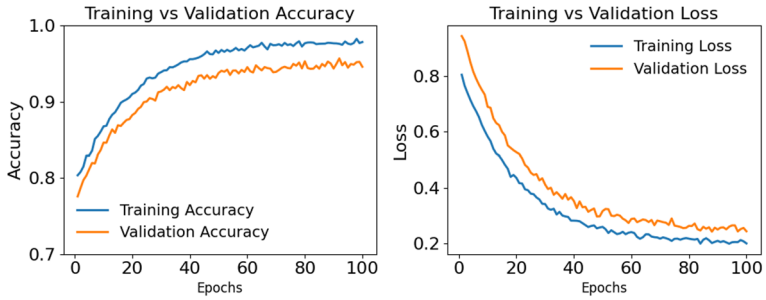


Fig. 4. Training and validation curves showing accuracy and loss progression across epochs.

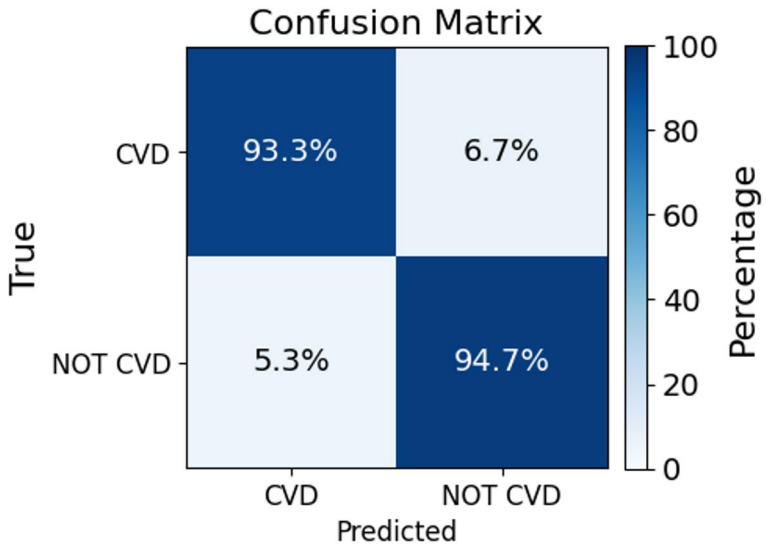


Fig. 5. Confusion matrix depicting the classification outcomes for cardiovascular disease and non-disease cases.

cases correctly identified and 6.7% misclassified. Focusing on the negative cases, there was also 94.7% accurate recognition, with the remaining 5.3% incorrectly labeled as positive. The combination of high true positive and true negative rates indicates that the system has a strong ability to reduce both false alarms and missed diagnoses, which is an important factor for clinical decision-making. The distribution of outcomes highlights the system's performance in terms of generalization and diagnostic accuracy, which is key for reliable performance in real-world applications, as shown in Fig. 5.

The results of the performance metrics clearly show that the proposed framework is significantly predictive and stable for various performance evaluation criteria. The accuracy and sensitivity are 94.98% each, indicating that not only is the model highly accurate but it is also able to identify real cardiovascular disease cases with very little loss of information. The specificity was 93.38%, which indicates the system's ability to reduce false alarms and avoid the misclassification of non-disease cases. The metric with the highest precision was the one with the lowest false-positive rate, reaching 95.43%, a good level of precision, especially for clinical reliability. The F1-score of 94.74% reinforces this balance of sensitivity and precision, attesting to the framework's ability to deliver reliable results across complementary measures. Overall, these findings show the strength and clinical feasibility of the method, which also turned out to be more promising in terms of disease detection and non-disease recognition, as demonstrated in Fig. 6.

The performance comparison across multiple algorithms highlights the superiority of the proposed framework over both traditional machine learning and deep learning models. Classical methods such as SVM, KNN, and Naive Bayes show moderate predictive strength, with accuracy values ranging from 82.9% to 86.1%, and sensitivities peaking at 88.3%. Deep learning approaches like LSTM and CNN provide slightly better outcomes, with accuracies of 86.95% and 87.05% respectively, and CNN achieving the highest sensitivity among baselines at 91.1%. However, these improvements remain limited when contrasted with the proposed system, which records an accuracy of 94.98%, sensitivity of 94.98%, specificity of 93.38%, precision of 95.43%, and F-measure of 94.74. The balanced enhancement across all metrics demonstrates that the proposed model not only improves the ability to correctly identify cardiovascular disease cases but also reduces false positives, achieving a level of robustness and reliability that the baseline algorithms could not match, as clearly evidenced in Table 3.

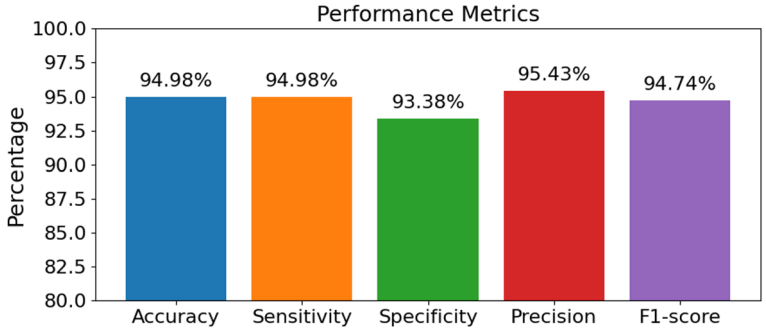


Fig. 6. Comparative performance metrics of the proposed model.

Algorithms	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F-measure (%)
SVM	82.9	85.8	81.75	84.3	84.9
KNN	84.65	87	82.6	86.4	86.25
Naive Bayes	86.1	88.3	83.2	87.15	86.9
LSTM	86.95	89.5	85.4	88.5	88.45
CNN	87.05	91.1	85.7	89.25	89.1
Proposed	94.98	94.98	93.38	95.43	94.74

Table 3. Comparative performance analysis of existing algorithms and the proposed model.

Algorithms	FPR (%)	FNR (%)	TNR (%)	RMSE (%)	MAE (%)
SVM	18.8	23.0	86.2	0.49	0.42
KNN	17.6	21.5	87.35	0.46	0.39
Naive Bayes	16.5	20.2	88.17	0.45	0.37
LSTM	16.0	18.9	88.91	0.42	0.34
CNN	15.2	14.0	89.11	0.37	0.31
Proposed	6.0	3.8	89.34	0.34	0.26

Table 4. Comparative error analysis.

To facilitate the comparison of different baseline classification models, namely SVM, KNN, Naïve Bayes, CNN, and LSTM, the same processing pipeline and multimodal feature representation of the proposed model were employed. This included ensuring that the same preprocessing steps, such as normalization, removal of outliers, anomaly adjustment, and standardization of features, were applied to all samples. Furthermore, to mitigate biases that can be introduced when using inconsistent features for model evaluation, the same multimodal feature representation, after SSECT feature extraction and ASO feature optimization, was used in all baseline classifiers. The parameters for each model were tuned through iterations based on the aim of stability and reliability of the prediction. The neighborhood size and kernel parameters were determined through experiments using conventional classifiers such as KNN, SVM, CNN, and LSTM models, and were optimized using the same optimization method.

The error-based evaluation reveals clear distinctions between traditional models, deep learning approaches, and the proposed framework. Classical techniques such as SVM, KNN, and Naive Bayes exhibit relatively high false positive rates ranging from 16.5% to 18.8% and false negative rates extending up to 23.0%, which indicates a tendency to misclassify both positive and negative cases. Deep learning models like LSTM and CNN demonstrate improvements, with CNN reducing the false negative rate to 14.0% and attaining lower error magnitudes reflected in RMSE of 0.37 and MAE of 0.31. Despite these advancements, the proposed method significantly outperforms all baselines by achieving the lowest false positive rate of 6.0% and false negative rate of 3.8%, while also maintaining the smallest RMSE (0.34) and MAE (0.26). Additionally, the true negative rate is slightly higher at 89.34%, which indicates a strong ability to correctly reject non-disease cases. These results underscore the robustness and efficiency of the proposed model in minimizing both types of classification errors and enhancing prediction stability, as demonstrated in Table 4.

The comparative analysis of error rate, agreement statistics, signal strength, and computational efficiency shows that the proposed framework clearly outperforms both conventional and deep learning models. Traditional approaches such as SVM, KNN, and Naive Bayes yield higher balanced error rates between 7.1% and 8.5%, while also presenting lower kappa scores that suggest weaker agreement with ground truth labels. Deep learning

Model	BER (%)	Kappa (%)	SNR (dB)	Execution time (s)
SVM	8.5	74.2	17.8	1.95
KNN	7.9	76.5	18.4	1.62
Naive Bayes	7.1	79.0	19.6	2.40
LSTM	6.8	81.2	20.1	1.30
CNN	5.9	84.0	21.5	3.05
Proposed	3.4	90.8	24.2	2.10

Table 5. Comparative evaluation of error rate, agreement, signal-to-noise ratio, and execution time across models.

Metric	Mean (%)	SD	SE	95% CI (lower–upper)
Accuracy	94.98	0.52	0.16	94.64–95.32
Sensitivity	94.98	0.60	0.19	94.60–95.36
Specificity	93.38	0.71	0.22	92.92–93.84
Precision	95.43	0.58	0.18	95.07–95.79
F1-score	94.74	0.55	0.17	94.40–95.08

Table 6. Statistical evaluation of performance metrics with mean, variability, and confidence intervals.

models improve these results, with CNN achieving a kappa value of 84.0% and an SNR of 21.5 dB, though at the cost of longer execution time of 3.05 s. In contrast, the proposed method achieves the lowest balanced error rate of 3.4% alongside the highest kappa score of 90.8% and the strongest SNR at 24.2 dB, confirming its superior reliability and predictive strength. Furthermore, its execution time of 2.10 s demonstrates that this performance advantage is attained without excessive computational burden, making it a practical as well as accurate solution for clinical decision support, as evidenced in Table 5.

The computation time presented in this work indicates the mean time needed to process one patient sample according to the methodology adopted in the work. This time involved preprocessing, feature extraction, feature fusion, and cardiovascular risk classification using DIR-GAN. The time taken was recorded by noting down the initial and end system times for processing one sample, then taking the average of several processing sessions. All experiments were performed using a computer with Intel® Core™ i9-12900K CPU, 64 GB of DDR5 RAM, and NVIDIA® RTX 4090 GPU with 24 GB of VRAM running on Python 3.10 & PyTorch 2.0.1.

The statistical evaluation confirms that the proposed framework achieves highly reliable performance with minimal variability across key classification measures. The mean accuracy is reported at 94.98% with a narrow standard deviation of 0.52, showing the model's ability to maintain stable outcomes across multiple trials. Sensitivity and specificity follow similar trends, with sensitivity at 94.98% and specificity at 93.38%, each supported by tight confidence intervals that demonstrate consistency in detecting both positive and negative cases. Precision is slightly higher at 95.43%, reflecting the model's strong ability to minimize false positives, while the F1-score of 94.74% validates its balanced treatment of precision and recall. The small standard errors across all metrics further indicate that random fluctuations had negligible influence on the reported results, highlighting the framework's robustness and reproducibility. Collectively, these findings emphasize that the model delivers reliable, generalizable predictions suitable for clinical application, as presented in Table 6.

The ROC curve provides a comprehensive view of the model's discriminative ability, illustrating its effectiveness in separating positive and negative cases across varying thresholds. The steep rise of the curve toward the top-left corner indicates that the classifier maintains a high true positive rate while keeping false positives relatively low. The computed AUC value of 0.947 demonstrates excellent classification power, signifying that the framework is capable of making highly reliable predictions in distinguishing between cardiovascular and non-cardiovascular cases. At the chosen operating point, the model achieved a true positive rate of 93.3% and a true negative rate of 89.3%, showing a well-balanced trade-off between sensitivity and specificity. This strong ROC performance confirms the robustness and clinical applicability of the proposed method in practical diagnostic settings, underscoring its suitability for risk assessment tasks as evidenced in Fig. 7.

The results of the five-fold cross-validation demonstrate that the proposed framework delivers highly consistent and stable performance across all evaluation metrics. Accuracy values remain tightly clustered between 94.70% and 95.10%, indicating minimal fluctuation in predictive capability across different folds. Precision ranges from 95.10% to 95.60%, showing that the model consistently minimizes false positives, while F-measure values between 94.60% and 95.00% confirm its balanced treatment of precision and recall. Similarly, specificity maintains a narrow range of 93.10% to 93.50%, reflecting the ability to reliably identify non-disease cases, and sensitivity remains strong between 94.70% and 95.10%, highlighting its reliability in detecting true cases of disease. The small variations across folds suggest that the model avoids overfitting and generalizes well to unseen data, reinforcing its robustness and clinical applicability as presented in Table 7.

The ablation work clearly demonstrates the contribution of each component to the overall performance of the proposed framework. When ABF is excluded, accuracy drops to 93.20%, showing that noise reduction and edge

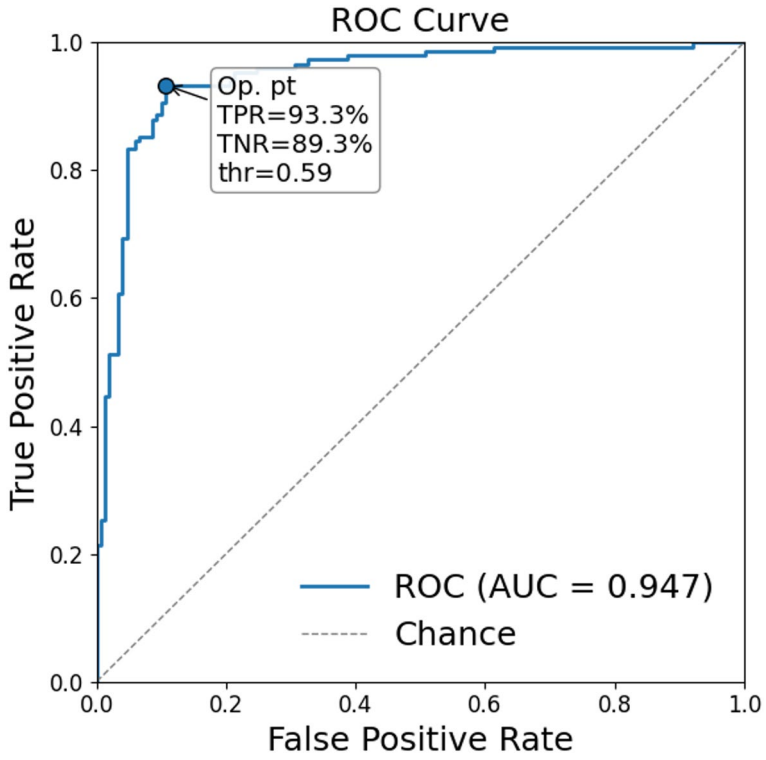


Fig. 7. ROC curve.

Fold	Accuracy (%)	Precision (%)	F-measure (%)	Specificity (%)	Sensitivity (%)
1	94.70	95.10	94.60	93.10	94.70
2	95.10	95.60	95.00	93.50	95.10
3	94.80	95.40	94.70	93.30	94.90
4	95.00	95.50	94.90	93.40	95.00
5	94.90	95.30	94.80	93.40	94.90

Table 7. Five-fold cross-validation results.

No	ABF	KDFCM	SSECT	DIR-GAN	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F1-score (%)
1	✗	✓	✓	✓	93.20	93.05	91.80	93.90	93.10
2	✓	✗	✗	✗	90.25	90.10	88.85	91.00	90.15
3	✓	✓	✓	✗	92.55	92.40	91.05	92.80	92.20
4	✓	✓	✗	✓	93.40	93.20	91.90	94.00	93.00
5	✓	✓	✓	✓	94.98	94.98	93.38	95.43	94.74

Table 8. Ablation work demonstrating the impact of ABF, KDFCM, SSECT, and DIR-GAN on model performance.

preservation play a critical role in improving image quality before segmentation. Similarly, removing KDFCM or SSECT leads to further reductions, highlighting the importance of precise myocardial segmentation and multi-scale texture analysis in extracting informative imaging features. The absence of the GAN-based classifier reduces performance as well, proving its effectiveness in handling class imbalance and enhancing generalization. Individually, the partial configurations achieve reasonable accuracy between 90.25% and 93.40%, but they do not match the strength of the complete pipeline. When all four components—ABF, KDFCM, SSECT, and DIR-GAN—are integrated, the framework achieves its peak performance with an accuracy of 94.98%, sensitivity of 94.98%, specificity of 93.38%, precision of 95.43%, and F1-score of 94.74, confirming that the synergy of all modules is essential for optimal predictive capability as presented in Table 8.

References	Dataset type	Methodology	Accuracy (%)	Strength	Limitation
Tiwari et al.[2]	Cardiac CT	CNN-based classification	92.34	Good predictive performance	Limited multimodal integration
Theerthagiri et al.[4]	Clinical data	Machine learning model	88.84	Efficient classification	Limited robustness
Li et al.[7]	Imaging data	Deep learning framework	87.30	High specificity	Reduced multimodal representation
Gupta et al.[11]	Imaging + Clinical	Deep learning model	93.23	Improved predictive accuracy	Limited adversarial learning
Hasan et al.[15]	Clinical data	Conventional classification	73.74	Simple implementation	Low prediction performance
Proposed	Cardiac CT + Clinical Data	DIR-GAN + ASO-based feature fusion	94.98	Multimodal integration, adversarial robustness, improved feature discrimination	Higher computational complexity during training

Table 9. Performance comparison of existing methods and the proposed approach.

Optimization algorithm	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F-measure (%)
Genetic Algorithm (GA)	89.42	90.15	87.84	89.76	89.58
Particle Swarm Optimization (PSO)	91.08	91.94	89.63	91.52	91.28
Grey Wolf Optimization (GWO)	92.67	93.25	91.54	92.83	92.96
Whale Optimization Algorithm (WOA)	93.41	93.86	92.05	94.12	93.74
Adaptive Starfish Optimization (ASO) (Proposed)	94.98	94.98	93.38	95.43	94.74

Table 10. Comparative analysis of optimization algorithms for feature selection.

Processing stage	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F-measure (%)
Raw data (without preprocessing)	81.42	82.35	79.84	81.95	81.76
After data preprocessing	86.73	87.52	85.18	86.94	86.82
After feature extraction (SSECT)	91.64	92.13	90.26	91.88	91.75
After ASO-based feature selection and fusion	93.42	93.87	92.16	94.05	93.96
Proposed DIR-GAN framework	94.98	94.98	93.38	95.43	94.74

Table 11. Impact of preprocessing and feature extraction on model performance.

An ablation work conducted on Table 8 gives a clearer understanding regarding the contribution and interaction of individual modules. Removal of ABF led to decreased performance in terms of increased noise in images and loss of structural information for myocardial regions. Likewise, absence of KDFCM segmentations made myocardial localization worse, hence decreasing the discriminative power of imaging features. Failure to extract features using SSECT led to poor performance as well since multi-scale texture and frequency characteristics for myocardial tissues could not be captured well. Lastly, use of conventional classification instead of DIR-GAN showed decreased predictive robustness because of increased likelihood of errors due to misclassification of heterogeneously post-COVID patterns. It is, therefore, clear that each module interacts positively with other modules in order to achieve synergistic enhancement of overall performance in predicting cardiovascular risks.

The comparative evaluation for the proposed framework versus the other state-of-the-art methods is presented in Table 9 in order to evaluate the efficacy of the framework in predicting the risk of developing cardiovascular problems and their detection based on post-COVID analysis. The factors taken into account include the dataset employed, methodology adopted, predictive accuracy, pros, and cons in order to make a comprehensive evaluation of existing methodologies. It can be observed from the results that the proposed DIR-GAN model was able to achieve better predictive accuracy by integrating the multimodal approach involving the use of cardiac CT scans and clinical features while enhancing the accuracy of the features using adversarial learning and ASO.

Comparison of the methods of feature selection has been carried out, to determine the efficiency of the Adaptive Starfish Optimization (ASO) technique relative to other optimization techniques, as illustrated in Table 10. From the findings obtained, ASO exhibited superior performance in terms of predictability using all the metrics used. This is due to the efficient exploration-exploration balance mechanism of ASO, thus making it capable of selecting important features, including imaging and clinical markers, in the cardiovascular prediction for post-COVID-19 patients. It is noteworthy that while ASO is based on the conventional optimization technique, its significance in the current research work can be determined from the efficient incorporation of ASO in the multimodal prediction of cardiovascular conditions.

Quantitative results for the effects brought about by the preprocessing and feature extraction steps can be found in Table 11 below. It can be observed from the results that the data consistency was enhanced along

Feasibility aspect	Observation	Practical implication
Data availability	Uses cardiac CT and routine clinical biomarkers	Compatible with hospital diagnostic workflow
Computation time	2.10 s execution time	Supports near real-time clinical decision support
Hardware requirement	Intel i9 + RTX 4090	Deployable in research/clinical computing environments
Model robustness	Multimodal fusion + adversarial learning	Improved reliability under heterogeneous patient data
Risk stratification	Low, Moderate, High-risk classification	Clinically interpretable decision support

Table 12. Feasibility analysis of the proposed DIR-GAN framework.

Model	MAE (%)	MSE (%)	RMSE (%)
SVM	17.10	14.65	12.10
KNN	15.35	13.08	11.44
Naïve Bayes	13.90	11.52	10.73
LSTM	13.05	10.84	10.41
CNN	12.95	10.52	10.26
Proposed DIR-GAN	5.02	4.18	6.46

Table 13. Error-based performance analysis of the proposed framework.

Data representation	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F-measure (%)
Clinical data only	86.72	87.34	85.61	86.95	86.95
CT imaging only	90.83	91.15	89.74	90.92	90.94
Multimodal fusion (Proposed)	94.98	94.98	93.38	95.43	94.74

Table 14. Comparative performance analysis across different data representations.

CNN architecture	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F-measure (%)
LeNet	84.76	86.42	83.15	85.74	86.02
AlexNet	87.92	89.35	86.44	88.63	88.98
VGG16	89.84	91.08	88.26	90.43	90.75
ResNet50	91.75	92.43	90.28	91.84	92.13
DenseNet121	92.68	93.15	91.62	93.06	93.10
CNN (Baseline)	87.05	91.10	85.70	89.25	89.10
Proposed DIR-GAN	94.98	94.98	93.38	95.43	94.74

Table 15. Comparative performance analysis using different CNN architectures.

with noise reduction during preprocessing, while the feature extraction method was effective in representing myocardial anomalies through discriminative representation.

The feasibility of the framework is shown in Table 12 to determine how feasible the framework is in terms of application to the prediction of cardiovascular risks among post-COVID-19 patients. It has been shown that the framework is indeed computationally feasible, clinically relevant, and can be incorporated into diagnostic processes owing to its efficient execution time and risk stratification.

Performance evaluation based on errors was done using MAE, MSE, and RMSE, as indicated in Table 13 below. These were used to measure prediction errors and classification consistency between different methods. The new DIR-GAN system gave the lowest error rates, implying better prediction reliability and minimized deviation in cardiovascular risk prediction compared to the traditional systems.

To assess the efficacy of the suggested model across heterogeneous data sources, a comparative result analysis was conducted employing clinical data, CT image data, and multimodal fusion data. Comparative assessment of results for accuracy, sensitivity, specificity, precision, and F-measure is tabulated in Table 14 indicate disparities in the predictability of the data. As can be seen from the results, multimodal fusion consistently yielded better results due to the integration of complementary information.

For further analysis on the efficiency of the suggested model, comparative performance analysis was conducted using various CNN models such as LeNet, AlexNet, VGG16, ResNet50, and DenseNet121, as shown in Table 15. From the comparison analysis, it can be concluded that deep CNN models outperformed classical

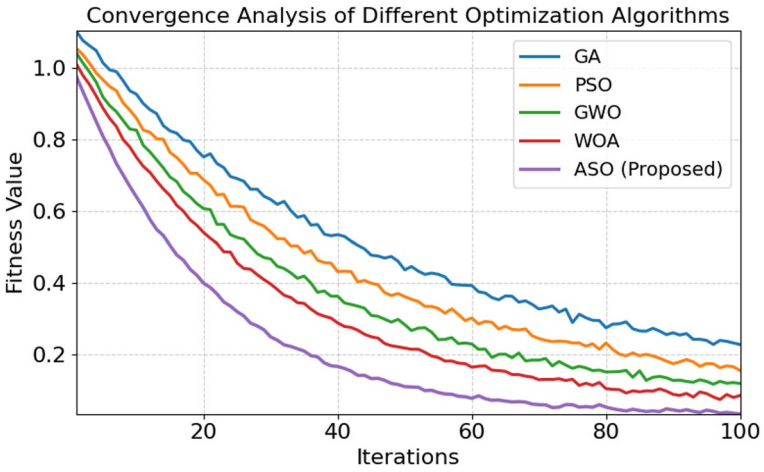


Fig. 8. Convergence analysis of different optimization algorithms.

Optimization algorithm	Iterations to convergence	Final fitness value	Convergence stability
GA	82	0.842	Moderate
PSO	75	0.871	Moderate
GWO	68	0.893	Good
WOA	61	0.912	Good
ASO (Proposed)	48	0.947	High

Table 16. Convergence analysis of optimization methods.

Module	Computational complexity
ABF	$O(nk)$
KDFCM	$O(nct)$
SSECT	$O(n\log n)$
ASO	$O(ptd)$
DIR-GAN	$O(ENW)$

Table 17. Computational complexity analysis of the proposed framework.

CNNs regarding prediction accuracy; however, the proposed DIR-GAN model performed even better compared to other models because of its unique characteristics of multimodal feature fusion and adversarial learning.

The convergence analysis for the various techniques was done for evaluating the stability of the optimization process and its convergence pattern while doing feature selection. This is shown in Fig. 8 and described in Table 16. The ASO technique that has been proposed shows better performance than traditional techniques as far as faster convergence with fewer iterations and better optimization stability.

Computational complexity of the proposed model has been considered to assess its scalability and feasibility for execution. Though the training procedure of the proposed DIR-GAN involves increased computational complexity owing to adversarial training, the inference speed achieved by the authors of 2.10 s shows that it is feasible enough for near real-time prediction of cardiovascular risks. Complexity contributions of different components of the proposed system are discussed in Table 17.

For enhancing the interpretability of clinical interpretation, SHAP-based feature importance analyses were done for the optimized set of clinical features, as depicted in Fig. 9. It can be observed from these results that the important clinical predictors in cardiovascular risk estimation include major vessel, thalassemia, ST-depression (Oldpeak), chest pain type, age, maximum heart rate, cholesterol, and resting blood pressure. Specifically, medically important thresholds like age greater than 60, cholesterol higher than 240 mg/dl, resting blood pressure associated with hypertension greater than 140 mmHg, and ST-depression (Oldpeak > 2.0) can be considered as significant indicators of high cardiovascular susceptibility.

Table 18 highlights some of the practical aspects of deploying the proposed DIR-GAN model for predicting the probability of having cardiovascular disease among post-COVID-19 patients. Besides the computational

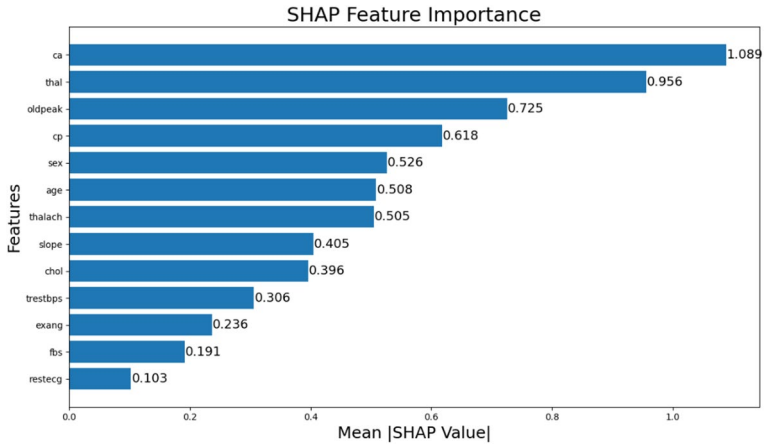


Fig. 9. SHAP-based clinical feature importance analysis for cardiovascular risk prediction.

Parameter	Value/Description
Training hardware	Intel i9-12900K, 64 GB RAM, RTX 4090 (24 GB VRAM)
Framework	Python 3.10, PyTorch 2.0.1
Average inference time per patient	2.10 s
Deployment memory requirement	Moderate GPU memory requirement
Prediction mode	Near real-time cardiovascular risk prediction
Resource-constrained optimization	Future lightweight deployment through pruning/quantization

Table 18. Practical deployment characteristics of the proposed DIR-GAN framework.

aspects associated with the training of the model, the table also highlights the computational costs involved when deploying the proposed system as well as ways of optimizing such deployment.

Other than the complexity of training the model, factors that would influence its deployment for actual clinical use were analyzed. From the results presented in Table 18, the proposed DIR-GAN framework took an average inference time of about 2.10 s per patient, making the tool suitable for near real-time cardiovascular risk prediction among post-COVID-19 patients. While the training of the model necessitated the use of high-end GPUs, the deployment phase did not involve significant computing requirements, making it feasible for computation in the workstation setup. Light-weighting approaches such as pruning, quantification, and simplified inference could be explored for improved deployment in resource-limited environments.

Discussion and limitations

The improved performance of the proposed DIR-GAN model in comparison to the CNN and LSTM baselines could be explained by the synergy between multimodal feature fusion and adversarial learning. To make sure that the evaluation was done fairly, all other baseline models, such as SVM, KNN, Naïve Bayes, LSTM, and CNN models, were analyzed by applying the preprocessing pipeline consisting of feature normalization, detection and removal of outliers, refinement of anomalies, and ASO-based feature selection in the case of fused imaging-clinical feature representation. Nonetheless, while traditional CNN and LSTM networks learn deterministic mappings based on features, the discriminator of DIR-GAN is trained in an adversarial environment where the generator provides it with new and complex variations of features. Such a design allows for improving decision boundaries, decreasing sensitivity to heterogeneous post-COVID conditions, and minimizing the misclassification rates by providing the network with more general and discriminative feature representations.

Apart from inference latency, the computational cost of implementing the proposed DIR-GAN architecture was also taken into account to ensure its potential utility in the clinical setting. The model training process was executed on a workstation powered by an Intel® Core™ i9-12900K CPU, 64 GB DDR5 RAM, and an NVIDIA® RTX 4090 graphics card (24 GB VRAM) using Python 3.10 and PyTorch 2.0.1 software. Even though the observed execution latency of 2.10 s shows that predictions are made quickly enough for real-time use, this work concentrates on the aspect of efficiency at the inference stage and model accuracy. Additionally, although the discriminator in the developed model is trained to predict patients' risk groups (low, moderate, and high), the current analysis was based on global performance measures. Class-wise evaluation will be considered in subsequent studies to validate model reliability.

While the framework is demonstrated with promising results for predicting post-COVID-19 disease status, there are some limitations that need to be noted. To begin with, the use of open-source data in the research implies that the work has limitations due to the heterogeneity of physiology and biomarkers in different people affected by post-COVID-19 symptoms. Another limitation is related to the lack of full demographic

information regarding the age, gender, and comorbidities in the imaging dataset, which can affect the results. Furthermore, while adversarial learning and multimodal feature integration enhanced the robustness of the predictions, the evaluation included aggregate classification performance but did not provide information on class-specific classification of moderate risk prediction cases. Finally, even though the sample size of the dataset was augmented, it is still relatively small, and the computational complexity of the DIR-GAN framework is increased during training. However, this is offset by the efficiency of inference-phase execution.

Further research can be conducted to optimize scalability and robustness of the proposed framework by including large, longitudinal, and more demographically diverse datasets. Moreover, a rigorous class-wise evaluation of moderate-risk cardiovascular disease prediction, use of lightweight DIR-GAN models for computational purposes, explainable AI models for better interpretability, and external validation of the clinical model in varied settings will be investigated.

The transformer-based architectures, especially Vision Transformer (ViT) and Swin Transformer, showed great promise in many recent medical imaging studies such as cardiac segmentation, CT image analysis, and multimodal feature learning, due to their superior ability to capture long-term dependencies in data. Nevertheless, the transformer architectures were not applied in the current work due to some practical issues in the dataset, especially in terms of the scale of the dataset, its multimodal heterogeneity, and computational efficiency. Generally, the transformers needed larger datasets and computational resources to learn more complex structures in the dataset and avoid overfitting, while in this work, the available public multimodal datasets were comparatively small and did not contain corresponding patient-level information. Consequently, the proposed DIR-GAN architecture was utilized to facilitate multimodal feature learning, discrimination, and stability under computational constraints. Meanwhile, the transformer architectures could be considered an interesting future direction of research. The comparisons between ViT- and Swin Transformer-based approaches and multimodal CV risk prediction frameworks would be carried out.

Despite the impressive predictive ability exhibited by the proposed methodological approach, certain considerations need to be made regarding its potential use within the healthcare setting. Specifically, the DIR-GAN proposed herein could serve as a decision-support tool for healthcare professionals in the process of predicting patient susceptibility to the development of CVD. The DIR-GAN model can be regarded as an alternative to the traditional clinical assessment of cardiovascular disease risk using population-based markers, such as the ASCVD risk score. In comparison with the ASCVD scoring technique that only accounts for population markers, the new approach utilizes myocardial imaging markers extracted from the results of cardiac CT scans in addition to cardiovascular factors. Nevertheless, the current investigation was not aimed at assessing the clinical potential of the method as no comparison to existing cardiovascular risk scores was conducted.

The suggested DIR-GAN model can be understood more as a domain-adaptation multimodal cardiovascular risk prediction framework than an entirely new generative adversarial network paradigm. The key point in the current work is to develop the idea of combining adversarial learning with multimodal cardiac CT imaging, ASO-based feature optimization, and structured cardiovascular biomarkers for estimating cardiovascular vulnerability after the pandemic infection with COVID-19. It is not the idea of creating a brand-new GAN model but the focus on methodological approaches, multimodal feature extraction, and cardiovascular risk prediction.

Although a five-fold cross-validation approach was used to enhance the robustness analysis process, cross hospital validation was not carried out because there were no available compatible post-COVID cardiovascular datasets with uniform multimodal annotations. This means that the proposed approach has been validated based on publicly available heterogeneous datasets and that care should be taken when evaluating its applicability. Moving forward, efforts will be made toward external validation.

Conclusion

This work offers a large and viable model that can be used to predict cardiovascular risk in patients with post-COVID-19. This strategy involves multimodal data (cardiac CT images and structured clinical records) that are integrated via a carefully designed pipeline. Starting from denoising the imaging data with Adaptive Bilateral Filtering, the system segments the data to obtain nuanced structural indicators of the disease through KDFCM and extracts the texture-frequency features of the data through SSECT. Simultaneously, the clinical features are carefully preprocessed, optimized, and reduced in dimension by Isolation Forest and Adaptive Starfish Optimization. Classification is performed synergistically between both modalities using feature-level fusion. The generator-discriminator architecture of the Deep Image Recognition–based Generative Adversarial Network (DIR-GAN) is used for classification. DIR-GAN is a classification network based on Deep Image Recognition (DIR) and Generative Adversarial Network (GAN) to improve feature generalization and classification accuracy. The framework achieved 94.98% accuracy, 95.43% precision, and 94.74% F1-score, outperforming traditional ML and deep learning models across all metrics, including error rates, sensitivity, and specificity. The low RMSE (0.34) and MAE (0.26), along with a Kappa score of 90.8%, confirm its reliability. Furthermore, the model exhibits computational efficiency, completing classification tasks in just 2.10 s, with robust five-fold cross-validation outcomes. Longitudinal clinical information (such as follow-up imaging and the evolution of biomarkers over time) can be added to the framework to improve risk trajectory prediction. The integration of wearable sensor data and real-time health monitoring could enable dynamic updates to patient risk profiles. The model will be further generalized when extended to a broader population and imaging modalities, such as cardiac MRI. Furthermore, the implementation of the framework as a clinical decision support system equipped with an interactive interface may present this research as a practical intervention in hospitals and provide personalized pathways of care for patients with cardiovascular complications in the post-COVID-19 era.

Data availability

The datasets analyzed in this work are publicly available from established open-access repositories and are fully anonymized. https://archive.ics.uci.edu/ml/datasets/Heart+Disease.

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References

1. Kanagarathinam, K. & Manikandan, R. Machine learning-based risk prediction model for cardiovascular disease using a hybrid dataset. *Data Knowl. Eng.* **140**, 102042 (2022).
2. Tiwari, A., Chugh, A. & Sharma, A. Ensemble framework for cardiovascular disease prediction. *Comput. Biol. Med.* **146**, 105624 (2022).
3. Khan, M. U. et al. Artificial neural network-based cardiovascular disease prediction using spectral features. *Comput. Electr. Eng.* **101**, 108094 (2022).
4. Theerthagiri, P. Predictive analysis of cardiovascular disease using gradient boosting based learning and recursive feature elimination technique. *Intell. Syst. Appl.* **16**, 200121 (2022).
5. Masumoto, A. et al. Impact of serum lactate dehydrogenase on the short-term prognosis of COVID-19 with pre-existing cardiovascular diseases. *J. Cardiol.* **4**, 501–508 (2022).
6. Sapra, V. et al. Integrated approach using deep neural network and CBR for detecting severity of coronary artery disease. *Alex. Eng. J.* **68**, 709–720 (2023).
7. Li, P., Yongmei, H. & Liu, Z.-P. Prediction of cardiovascular diseases by integrating multi-modal features with machine learning methods. *Biomed. Signal Process. Control* **66**, 102474 (2021).
8. Kumar, A. & Rekha, R. An improved hawks optimizer based learning algorithms for cardiovascular disease prediction. *Biomed. Signal Process. Control* **81**, 104442 (2023).
9. Sumwiza, K., Twizere, C., Rushingabigwi, G., Bakunzibake, P. & Bamurigire, P. Enhanced cardiovascular disease prediction model using random forest algorithm. *Inform. Med. Unlocked* **41**, 101316 (2023).
10. Kiran, S., Reddy, G. R. & Dorthi, K. A gradient boosted decision tree with binary spotted hyena optimizer for cardiovascular disease detection and classification. *Healthc. Anal.* **3**, 100173 (2023).
11. Gupta, A., Jain, V. & Singh, A. Stacking ensemble-based intelligent machine learning model for predicting post-COVID-19 complications. *New Gener. Comput.* **40**(4), 987–1007 (2022).
12. Kumar, A. & Rekha, R. A dense network approach with gaussian optimizer for cardiovascular disease prediction. *New Gener. Comput.* **4**, 859–878 (2023).
13. Khan, M. U. et al. A two-stage classification model integrating feature fusion for coronary artery disease detection and classification. *Multimed. Tools Appl.* **10**, 13661–13690 (2022).
14. Nelson, I., Annadurai, C. & Devi, K. N. Retracted article: An efficient AlexNet deep learning architecture for automatic diagnosis of cardio-vascular diseases in healthcare system. *Wirel. Pers. Commun.* **1**, 493–509 (2022).
15. Hasan, N. & Bao, Y. Comparing different feature selection algorithms for cardiovascular disease prediction. *Health Technol.* **11**(1), 49–62 (2021).
16. Baghdadi, N. A. et al. Advanced machine learning techniques for cardiovascular disease early detection and diagnosis. *J. Big Data* **10**(1), 144 (2023).
17. Higaki, A. et al. Predictive value of neutrophil-to-lymphocyte ratio for the fatality of COVID-19 patients complicated with cardiovascular diseases and/or risk factors. *Sci. Rep.* **1**, 13606 (2022).
18. Sai Krishna Reddy, V., Meghana, P., Subba Reddy, N. V. & Ashwath Rao, B. Prediction on cardiovascular disease using decision tree and Naïve Bayes classifiers. In *Journal of Physics: Conference Series* Vol. 2161, 012015 (IOP Publishing, 2022).
19. Sinha, N., Jangid, T., Joshi, A. M. & Mohanty, S. P. iCARDO: A machine learning based smart healthcare framework for cardiovascular disease prediction. arXiv preprint arXiv:2212.08022 (2022).
20. Rahim, A. et al. An integrated machine learning framework for effective prediction of cardiovascular diseases. *IEEE Access* **9**, 106575–106588 (2021).
21. Yaqoob, M. M., Nazir, M., Khan, M. A., Qureshi, S. & Al-Rasheed, A. Hybrid classifier-based federated learning in health service providers for cardiovascular disease prediction. *Appl. Sci.* **13**(3), 1911 (2023).
22. Rustam, F. et al. Incorporating CNN features for optimizing performance of ensemble classifier for cardiovascular disease prediction. *Diagnostics* **12**(6), 1474 (2022).
23. Balakrishnan, C. & Ambeth Kumar, V. D. IoT-enabled classification of echocardiogram images for cardiovascular disease risk prediction with pre-trained recurrent convolutional neural networks. *Diagnostics* **4**, 775 (2023).
24. Narasimhan, G. & Victor, A. Empirical analysis of predicting heart disease using diverse datasets and classification procedures of machine learning. *Ain Shams Eng. J.* **16**(8), 103470 (2025).
25. Shrivastava, P. K., Sharma, M. & Kumar, A. HCBiLSTM: A hybrid model for predicting heart disease using CNN and BiLSTM algorithms. *Meas. Sens.* **25**, 100657 (2023).
26. Uddin, K. M., Ripa, R., Yeasmin, N., Biswas, N. & Dey, S. K. Machine learning-based approach to the diagnosis of cardiovascular vascular disease using a combined dataset. *Intell. Based Med.* **7**, 100100 (2023).
27. Ozcan, M. & Peker, S. A classification and regression tree algorithm for heart disease modeling and prediction. *Healthc. Anal.* **3**, 100130 (2023).
28. Majumder, A. B. et al. Heart disease prediction using concatenated hybrid ensemble classifiers. *Algorithms* **12**, 538 (2023).
29. Gupta, I., Bajaj, A., Malhotra, M., Sharma, V. & Abraham, A. Heart disease prediction using a hybrid feature selection and ensemble learning approach. *IEEE Access* https://doi.org/10.1109/ACCESS.2025.3583757 (2025).
30. Doppala, B. P., Bhattacharyya, D., Chakkravarthy, M. & Kim, T. H. A hybrid machine learning approach to identify coronary diseases using feature selection mechanism on heart disease dataset. *Distrib. Parallel Databases* **1**, 20 (2023).
31. Arunachalam, S. K. & Rekha, R. A novel approach for cardiovascular disease prediction using machine learning algorithms. *Concurr. Comput. Pract. Exp.* **19**, e7027 (2022).
32. Saranya, K., Karthikeyan, U., Kumar, A. S., Salau, A. O. & Tin Tin, T. DenseNet-ABiLSTM: Revolutionizing multiclass arrhythmia detection and classification using hybrid deep learning approach leveraging PPG signals. *Int. J. Comput. Intell. Syst.* **1**, 33 (2025).
33. Rajagopal, R. & Ranganathan, V. Evaluation of effect of unsupervised dimensionality reduction techniques on automated arrhythmia classification. *Biomed. Signal Process. Control* **34**, 1–8 (2017).
34. Solanki, S., Chouhan, S. S., Patel, R. K., Singh, U. P. & Namdev, M. An approach for classification of brain tumor using fully connected deep convolutional neural network. In *2024 4th International Conference on Sustainable Expert Systems (ICSES)* 1306–1311 (IEEE, 2024).
35. Selot, N., Panwa, A., Shukla, A., Chouhan, S. S., Patel, R. K. & Solanki, S. Comparative analysis of convolutional neural network and support vector machine for the prediction of Alzheimer's disease. In *International Conference on Applied Technologies* 56–66 (Springer, 2024).

36. Siddiqui, E. A. & Patel, R. K. A novel Dense-Net deep neural network method for classify tuberculosis stages using chest X-ray images with enhanced features. In *2024 First International Conference on Data, Computation and Communication (ICDCC)* 737–743 (IEEE, 2024).
37. Wang, Z., Liu, Q. & Dou, Q. Contrastive cross-site learning with redesigned net for COVID-19 CT classification. *IEEE J. Biomed. Health Inform.* **10**, 2806–2813 (2020).
38. Xu, G.-X. et al. Cross-site severity assessment of COVID-19 from CT images via domain adaptation. *IEEE Trans. Med. Imaging* **1**, 88–102 (2021).
39. Yang, J., Soltan, A. A. & Clifton, D. A. Machine learning generalizability across healthcare settings: insights from multi-site COVID-19 screening. *NPJ Digit. Med.* **1**, 69 (2022).

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Conceptualization: S. Diana Juliet; Data curation: S. Diana Juliet; Investigation: J. Banumathi, J. Anix Joel Singh; Methodology: S. Diana Juliet; Project administration: J. Banumathi, J. Anix Joel Singh; Supervision: J. Banumathi, J. Anix Joel Singh; Validation: J. Banumathi, J. Anix Joel Singh; Writing – original draft: S. Diana Juliet.

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Declarations

Competing interests
The authors declare no competing interests.

Ethical approval and informed consent
This work did not involve direct interaction with human participants or the collection of new patient data. The imaging dataset was obtained from the publicly accessible COVID-19 CT Scan Dataset (https://www.kaggle.com/datasets/ahmedtronic/covid-19), while structured clinical records were obtained from the UCI Machine Learning Repository – Heart Disease Dataset (https://archive.ics.uci.edu/dataset/45/heart+disease). All datasets used were publicly available and fully anonymized; therefore, additional ethical approval and informed consent were not required in accordance with applicable institutional and national guidelines. According to the source dataset documentation, the diagnostic utility of the CT images was verified by a senior radiologist at Tongji Hospital during the COVID-19 outbreak, and the data were curated for medical research and diagnostic purposes under relevant institutional procedures. All methods in this work were carried out using de-identified data and in accordance with relevant guidelines and regulations.

Additional information
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